UNIVERSAL POOFER CONTROL

REV A.3 | Functional architecture | 2026-10-03

INTERFACE CONTRACT

8 grounded contacts; nominal 20 mA supplied by board.

Continuous +/-100V field faults; negative faults masked.

8 NO valve outputs:12 VDC, 1 A design load per channel.

Buffered host/expander chain; separate module power.

Physical role jumper; no MCU control of role/source.

HARDWARE SAFETY CONTRACT

NC loop open/disconnected sets hardware stop latch.

Loop restored does not re-energize retained outputs.

Fresh ARM edge and power-good required to enable.

GPIO18 = loop healthy; GPIO21 = hardware RUN feedback.

NC relay contacts auxiliary; valve connectors are NO.

02 | Processor and service interfaces

Processor, reset and programming

File: Controller.kicad_sch

03 | Power and analog references

Logic supplies and monitored startup

File: Power.kicad_sch

04 | Register chains

8 command bits / 16 captured input bits

File: Serial_IO.kicad_sch

05 | Buffered expansion interface

Buffered chain and physical module role

File: Expansion.kicad_sch

06 | Hardware stop latch and coil gate

Asynchronous stop / edge-controlled restart

File: Hardware_Stop.kicad_sch

07 | Contact receivers 1-2

Protected contact sensing

File: Input_Receivers_1.kicad_sch

08 | Contact receivers 3-4

Protected contact sensing

File: Input_Receivers_2.kicad_sch

09 | Contact receivers 5-6

Protected contact sensing

File: Input_Receivers_3.kicad_sch

10 | Contact receivers 7-8

Protected contact sensing

File: Input_Receivers_4.kicad_sch

11 | Wetting sources 1-2

20 mA source and field fault cutoff

File: Contact_Sources_1.kicad_sch

12 | Wetting sources 3-4

20 mA source and field fault cutoff

File: Contact_Sources_2.kicad_sch

13 | Wetting sources 5-6

20 mA source and field fault cutoff

File: Contact_Sources_3.kicad_sch

14 | Wetting sources 7-8

20 mA source and field fault cutoff

File: Contact_Sources_4.kicad_sch

15 | NC emergency-stop field loop

Protected NC loop and tail termination

File: Stop_Field.kicad_sch

16 | Valve outputs 1-4

NO valve contacts and release clamps

File: Relay_Outputs_1.kicad_sch

17 | Valve outputs 5-8

NO valve contacts and release clamps

File: Relay_Outputs_2.kicad_sch

18 | Analog supplies, spare units and debug port

Analog bypassing and defined spare units

File: Analog_Support.kicad_sch

19 | Safety logic supplies and unused gates

Safety logic bypassing and spare units

File: Logic_Support.kicad_sch

20 | Buffered current reference and rail sequencing

Stable shared wetting-current reference

File: Current_Reference.kicad_sch

21 | I2C expansion and power-off isolation

Shared A bus / isolated private MCU bus

File: I2C_Expansion.kicad_sch

22 | Expansion defaults and buffer supplies

Safe control defaults and bypass

File: Expansion_Bias.kicad_sch

23 | Hardware monitoring of 5V and raw 12V

Logic-powered raw supply-loss protection

File: Rail_Supervision.kicad_sch

24 | Stable emitter feedback for all wetting sources

Stable transistor-loop feedback

File: Source_Feedback_Filters.kicad_sch

PCB placement stage only: no tracks, vias or copper pours.

Routing, actual-valve tests and independent system safety qualification remain required.

Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/-100V faults
Universal Poofer Control | Engineering

Sheet: /
File: Universal_Poofer_Control.kicad_sch

Title: 01 | System architecture

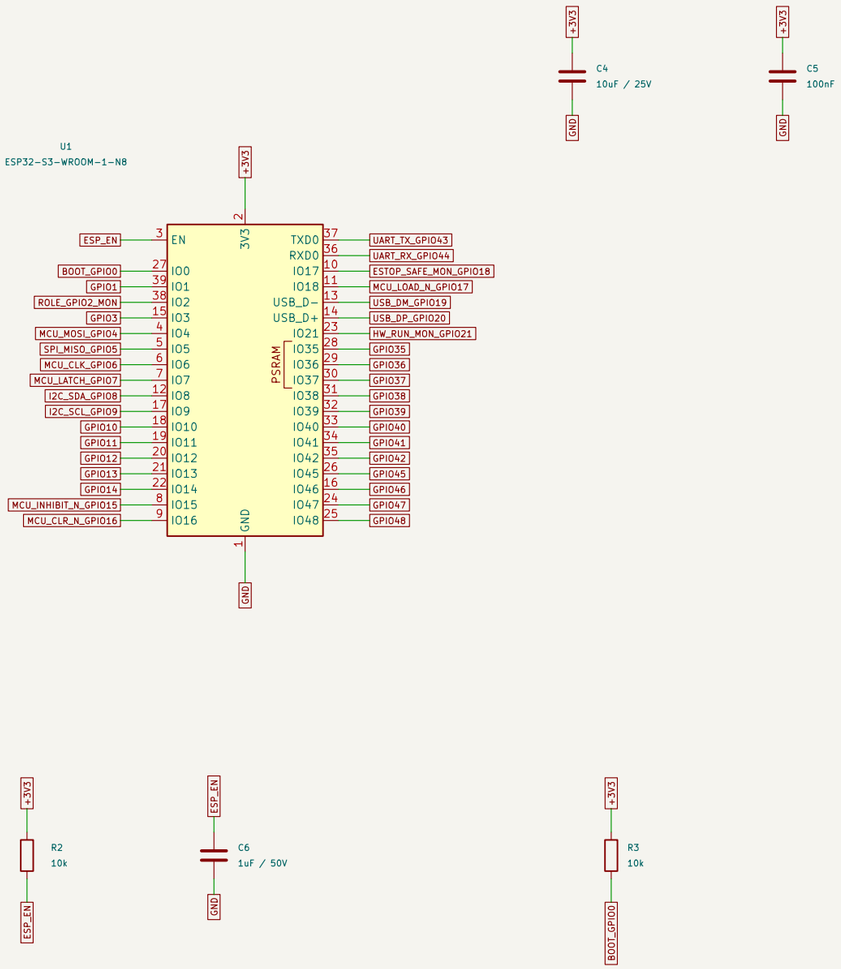
Size: A3	Date: 2026-10-03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 1/24

UNIVERSAL POOFER CONTROL | 02 | Processor and service interfaces

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

ESP32-S3 / RESET

WROOM-1-N8 | 8MB flash | no PSRAM



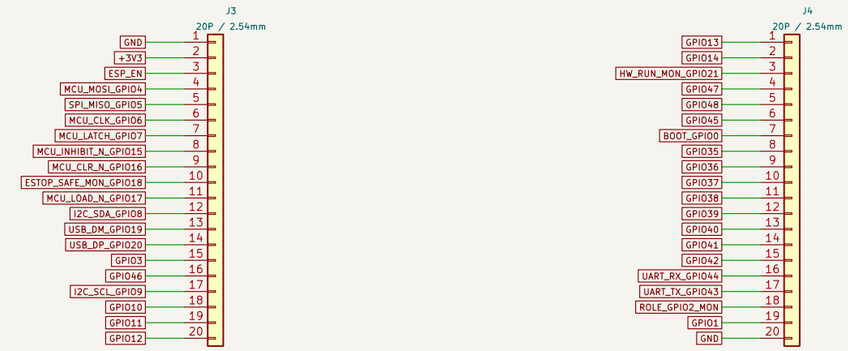
GPIO0/3/45/46 arestraps. NativeUSB pins exposed;useprotected short-wireadapter.

PROGRAMMING



GPIO BREAKOUTS

Module pin order; interface functions named



Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/-100V faults
Universal Poofer Control | Engineering

Sheet: /02 | Processor and service interfaces/
File: Controller.kicad_sch

Title: 02 | Processor and service interfaces

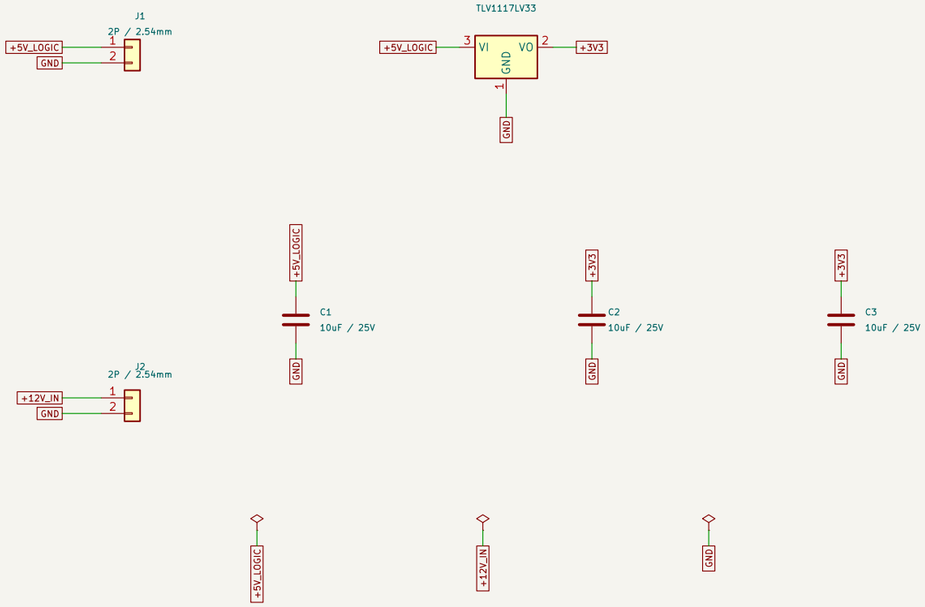
Size: A3	Date: 2026-10-03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 2/24

UNIVERSAL POOFER CONTROL | 03 | Power and analog references

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

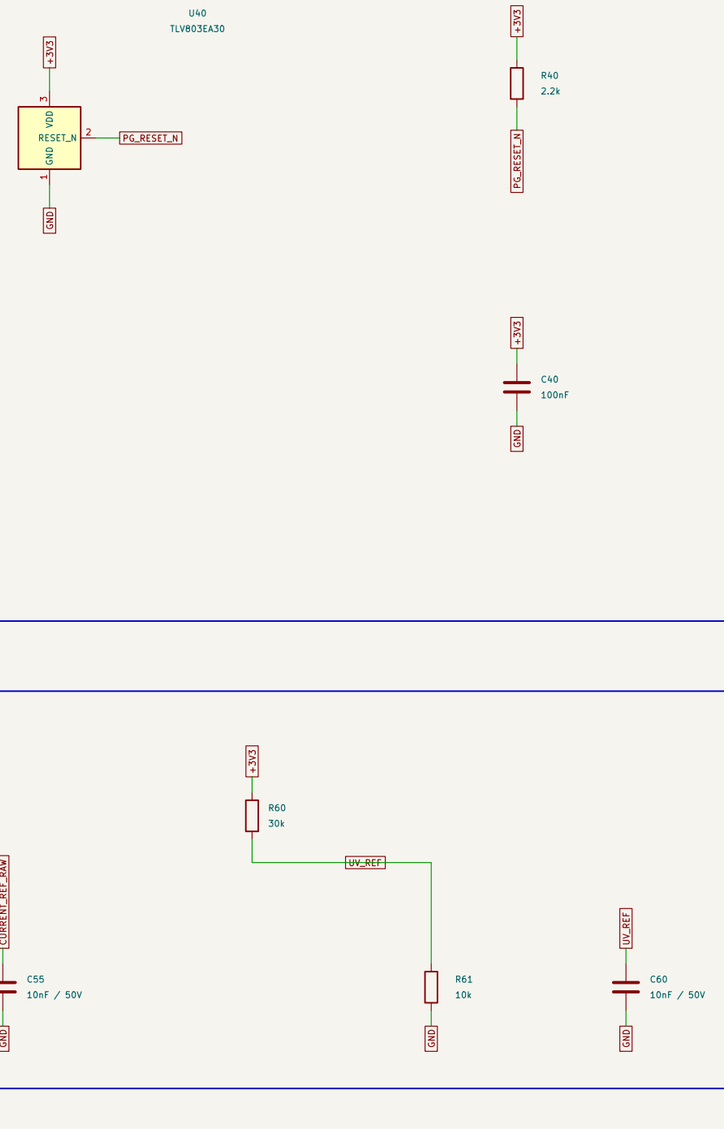
POWER INPUTS / 3.3V

5.0–5.25V>=1A;12V>=0.5A; independent modules

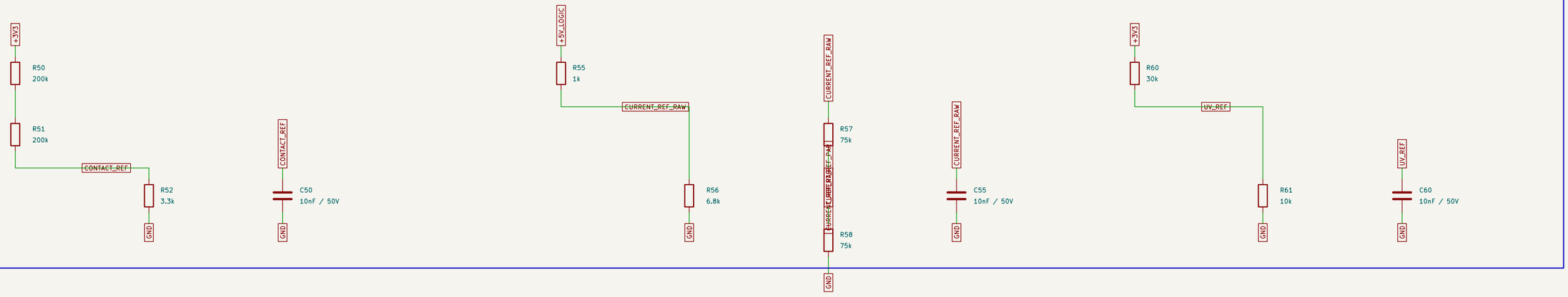


SUPERVISED STARTUP

3.08Vthreshold;130–270msresetdelay



SHARED REFERENCES



CURRENT_REF=5V x (6.8k||150k)/(1k+(6.8k||150k)); collector target-20mA.

Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/–100V faults
Universal Poofer Control | Engineering

Sheet: /03 | Power and analog references/
File: Power.kicad_sch

Title: 03 | Power and analog references

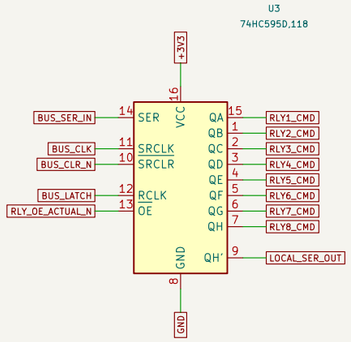
Size: A3	Date: 2026-10-03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 3/24

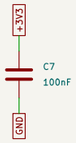
UNIVERSAL POOFER CONTROL | 04 | Register chains

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

595 RELAY COMMAND LATCH

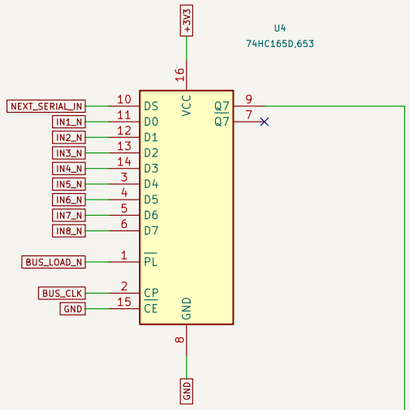
QA=valve1; latch commits afterfullshift

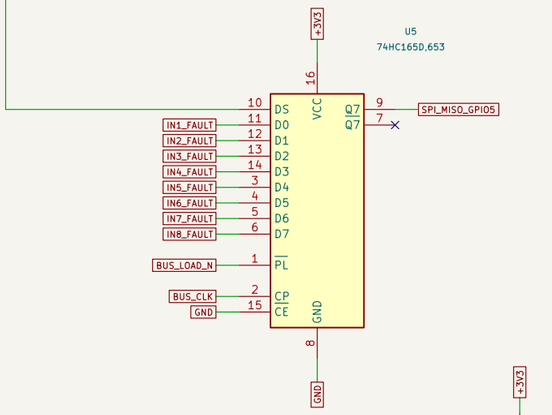


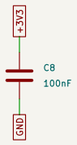


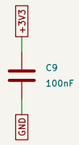
CONTACT / FAULT CAPTURE

16bits/module:FAULTthenCONTACTS_N









SampleQ7beforeclock. TailpulledHIGH fillsabsentmoduleswithFAULTFF,notfalseclosures.

Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/−100V faults
Universal Poofer Control | Engineering

Sheet: /04 | Register chains/
File: Serial_IO.kicad_sch

Title: 04 | Register chains

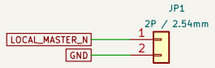
Size: A3	Date: 2026-10-03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 4/24

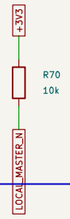
UNIVERSAL POOFER CONTROL | 05 | Buffered expansion interface

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

PHYSICAL MASTER / EXPANDER ROLE

GPIO2 monitors through100k; cannot override hardware jumper



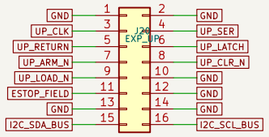


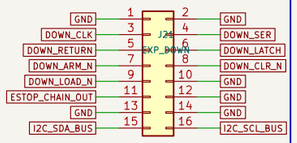




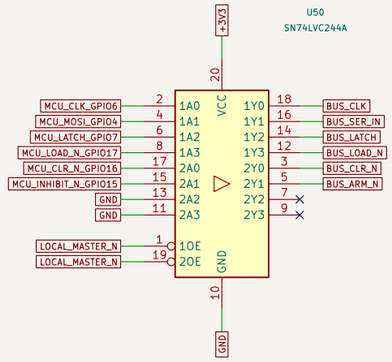
UP / DOWN PORTS

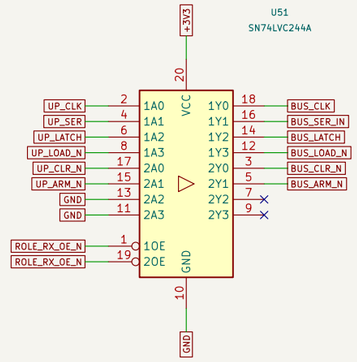
2x8male,2.54mm: independentmodulepower



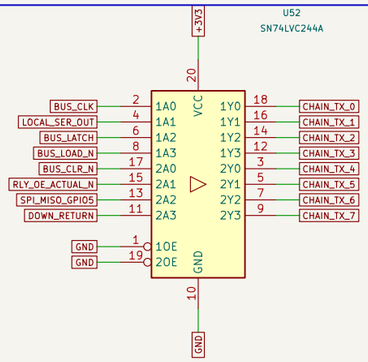


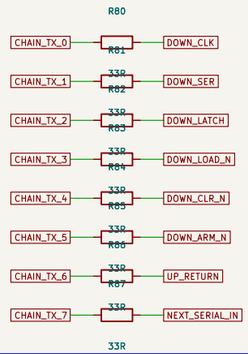
ISOLATED LOCAL / UP DRIVERS





REGENERATED CHAIN OUTPUTS





Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/−100V faults
Universal Poofer Control | Engineering

Sheet: /05 | Buffered expansion interface/
File: Expansion.kicad_sch

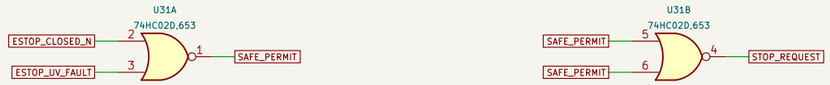
Title: 05 | Buffered expansion interface

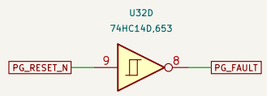
Size: A3	Date: 2026-10-03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 5/24

UNIVERSAL POOFER CONTROL | 06 | Hardware stop latch and coil gate

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

NC LOOP PERMISSION

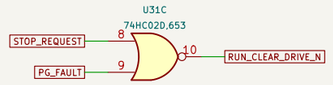


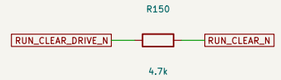


Arm clock is independent of permission: a held ARM cannot restart after stop restoration.

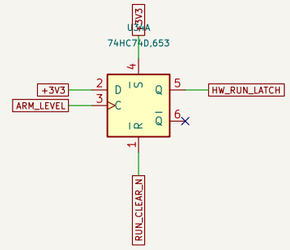
EDGE-ARMED HARDWARE LATCH

Loop restoration cannot clock the flip-flop

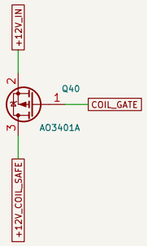


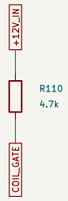




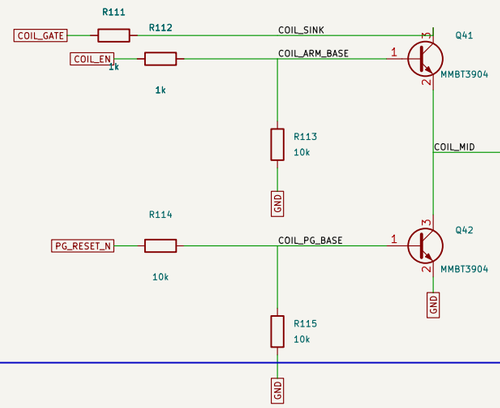


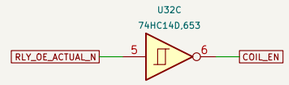
SUPERVISED COIL DISCONNECT

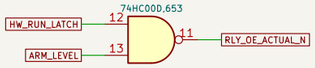












GPIO18 = loop permission; GPIO21 = hardware RUN. Fresh ARM edge clears a healthy stop.

External safety contactor must disconnect valve power if welded-contact protection is required.

Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/-100V faults
Universal Poofer Control | Engineering

Sheet: /06 | Hardware stop latch and coil gate/
File: Hardware_Stop.kicad_sch

Title: 06 | Hardware stop latch and coil gate

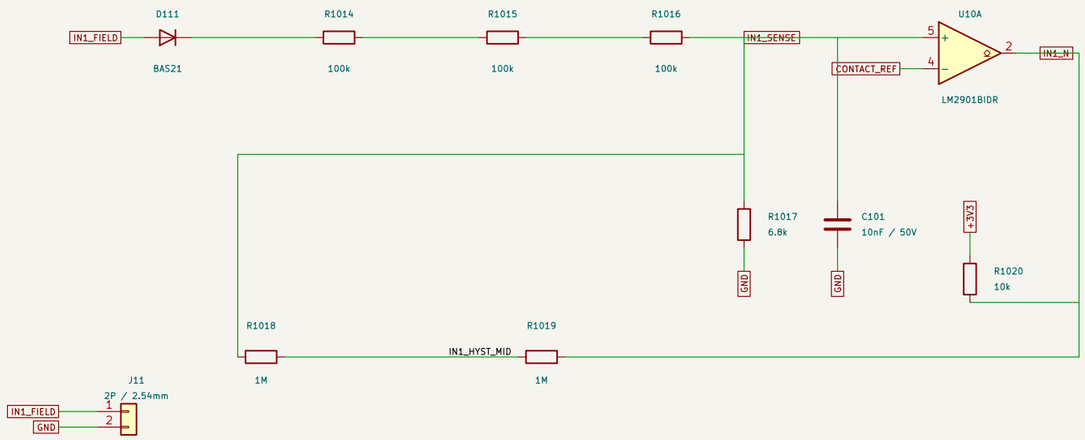
Size: A3	Date: 2026-10-03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 6/24

UNIVERSAL POOFER CONTROL | 07 | Contact receivers 1–2

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

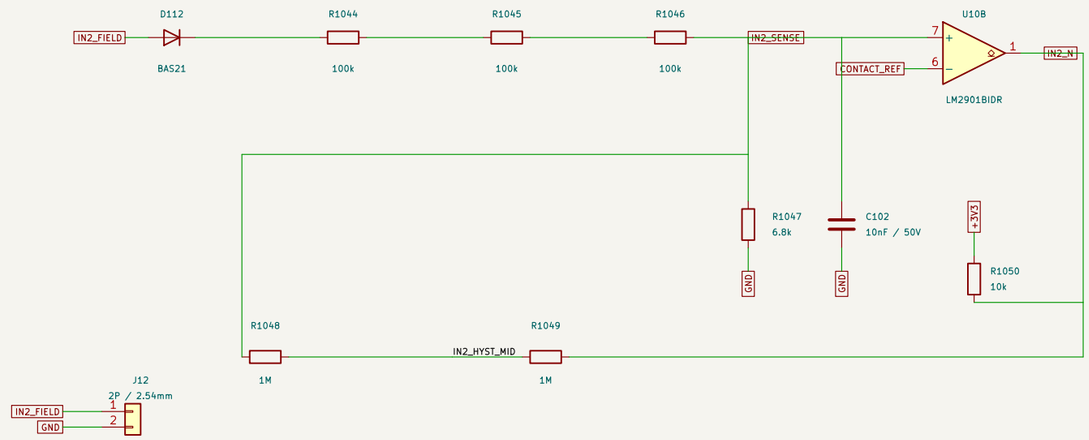
INPUT 1 | ACTIVE–LOW RECEIVER

LOW = closed; FAULT high masks invalid closure



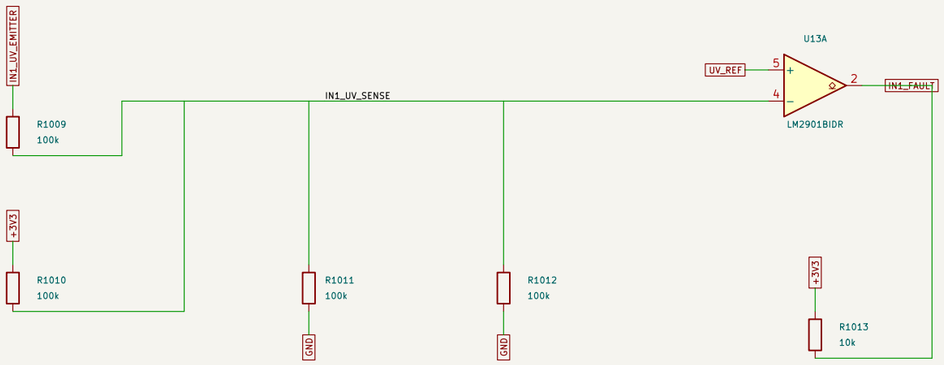
INPUT 2 | ACTIVE–LOW RECEIVER

LOW = closed; FAULT high masks invalid closure



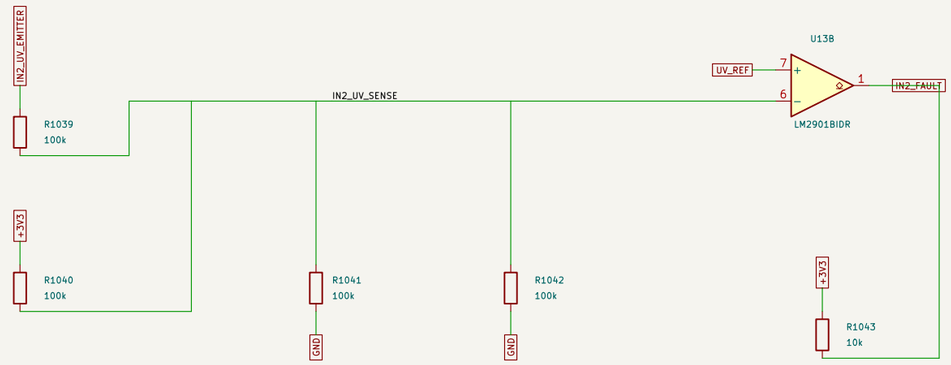
PROTECTED FAULT FLAG

Four 100k conductances bias and attenuate emitter



PROTECTED FAULT FLAG

Four 100k conductances bias and attenuate emitter



Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/–100V faults
Universal Poofer Control | Engineering

Sheet: /07 | Contact receivers 1–2/
File: Input_Receivers_1.kicad_sch

Title: 07 | Contact receivers 1–2

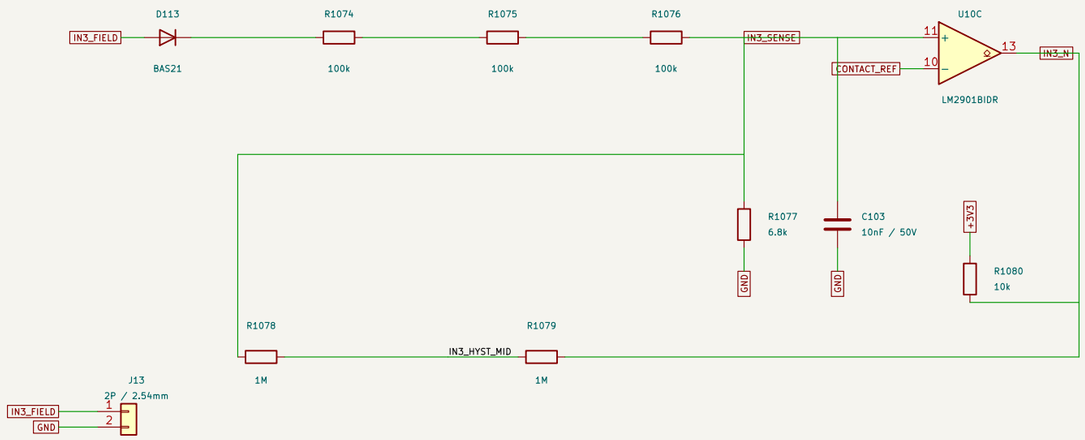
Size: A3	Date: 2026–10–03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 7/24

UNIVERSAL POOFER CONTROL | 08 | Contact receivers 3–4

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

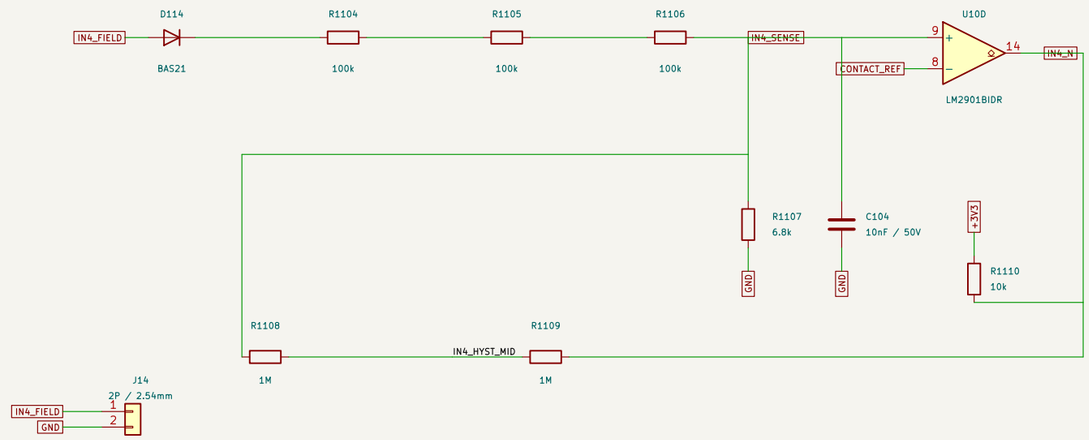
INPUT 3 | ACTIVE-LOW RECEIVER

LOW = closed; FAULT high masks invalid closure



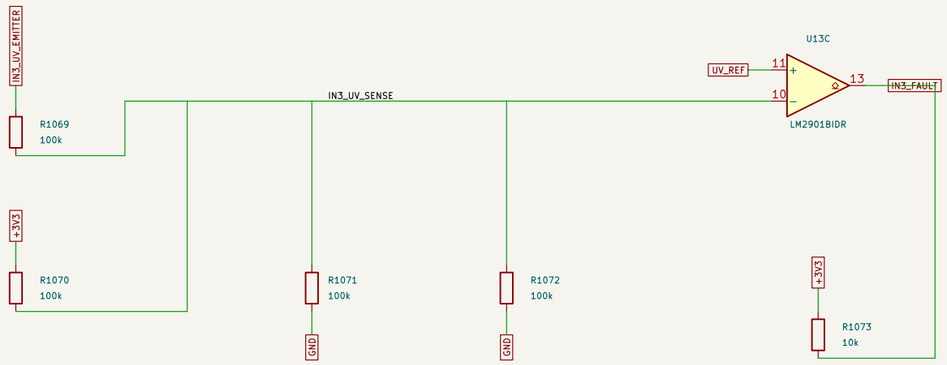
INPUT 4 | ACTIVE-LOW RECEIVER

LOW = closed; FAULT high masks invalid closure



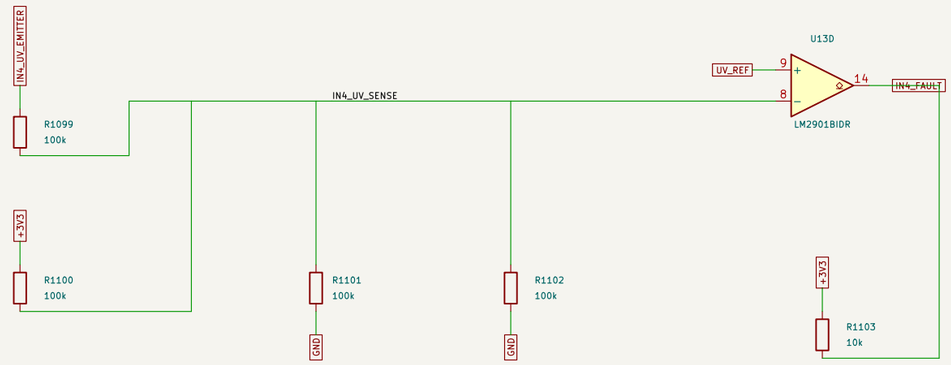
PROTECTED FAULT FLAG

Four 100k conductances bias and attenuate emitter



PROTECTED FAULT FLAG

Four 100k conductances bias and attenuate emitter



Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/−100V faults
Universal Poofer Control | Engineering

Sheet: /08 | Contact receivers 3–4/
File: Input_Receivers_2.kicad_sch

Title: 08 | Contact receivers 3–4

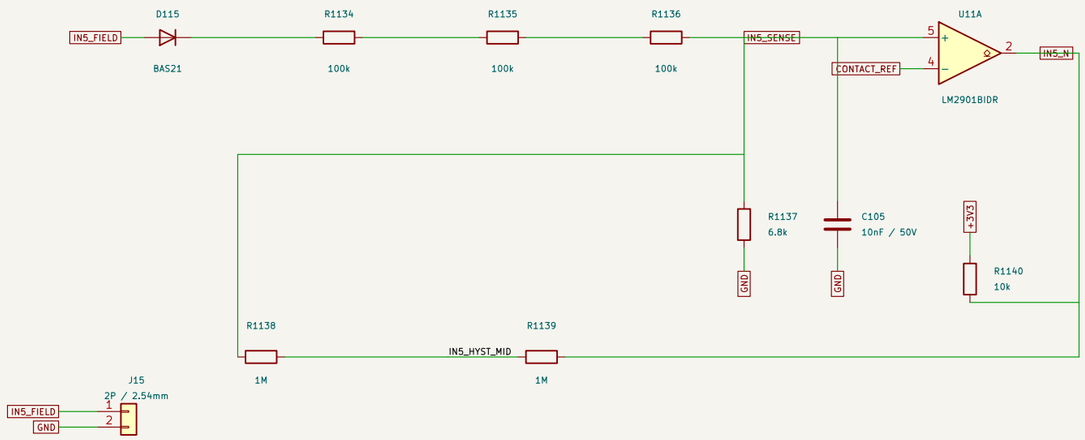
Size: A3	Date: 2026-10-03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 8/24

UNIVERSAL POOFER CONTROL | 09 | Contact receivers 5–6

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

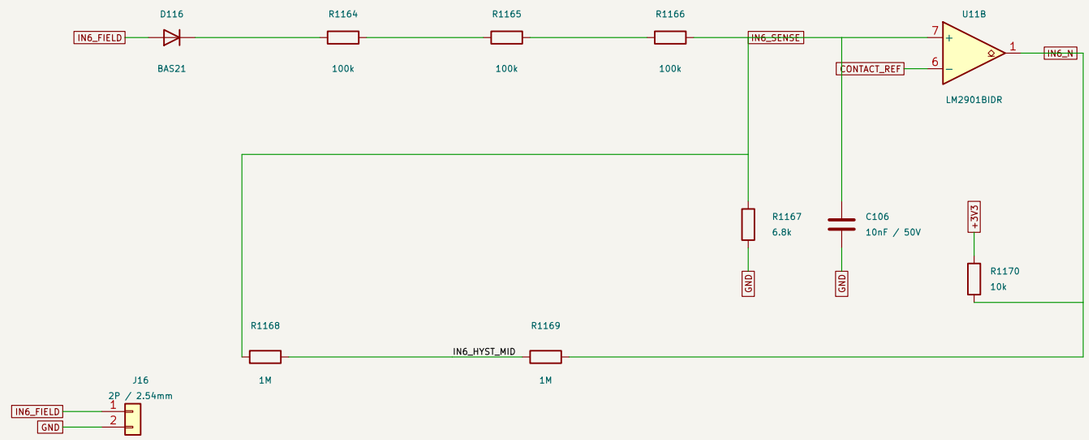
INPUT 5 | ACTIVE-LOW RECEIVER

LOW = closed; FAULT high masks invalid closure



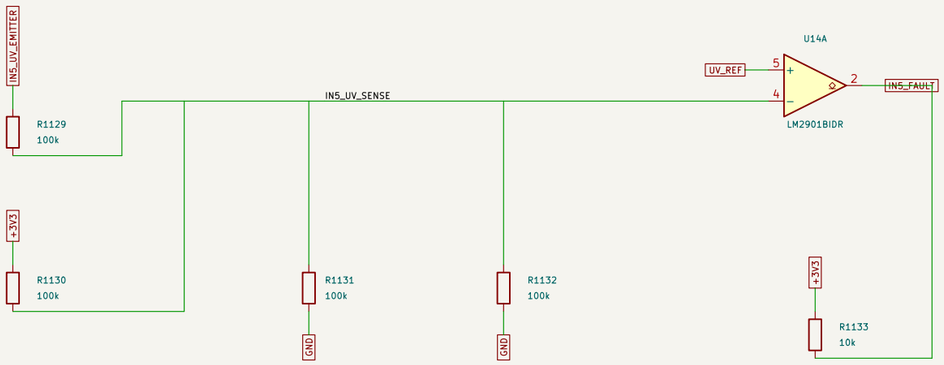
INPUT 6 | ACTIVE-LOW RECEIVER

LOW = closed; FAULT high masks invalid closure



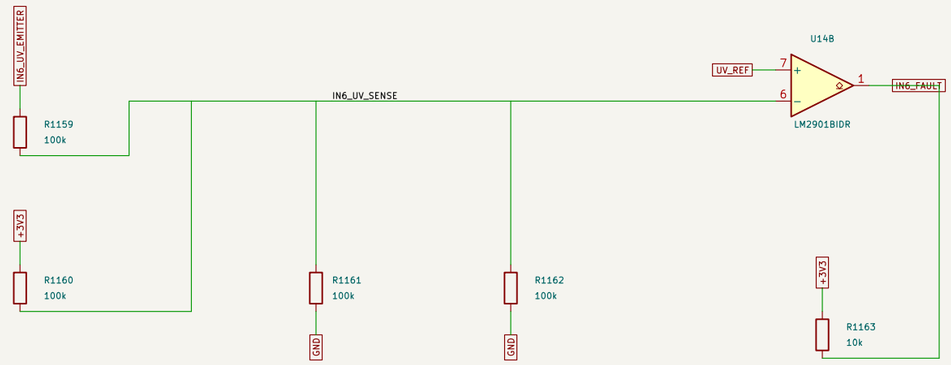
PROTECTED FAULT FLAG

Four 100k conductances bias and attenuate emitter



PROTECTED FAULT FLAG

Four 100k conductances bias and attenuate emitter



Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/−100V faults
Universal Poofer Control | Engineering

Sheet: /09 | Contact receivers 5–6/
File: Input_Receivers_3.kicad_sch

Title: 09 | Contact receivers 5–6

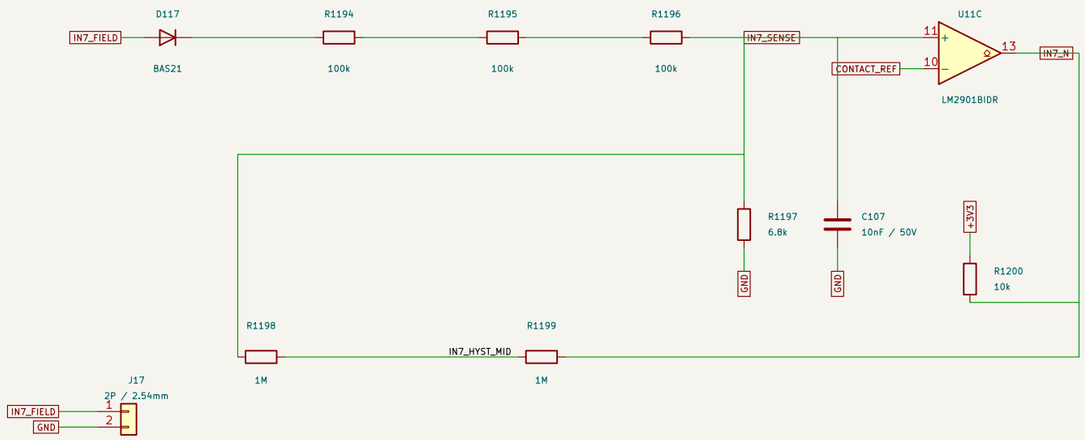
Size: A3	Date: 2026-10-03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 9/24

UNIVERSAL POOFER CONTROL | 10 | Contact receivers 7–8

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

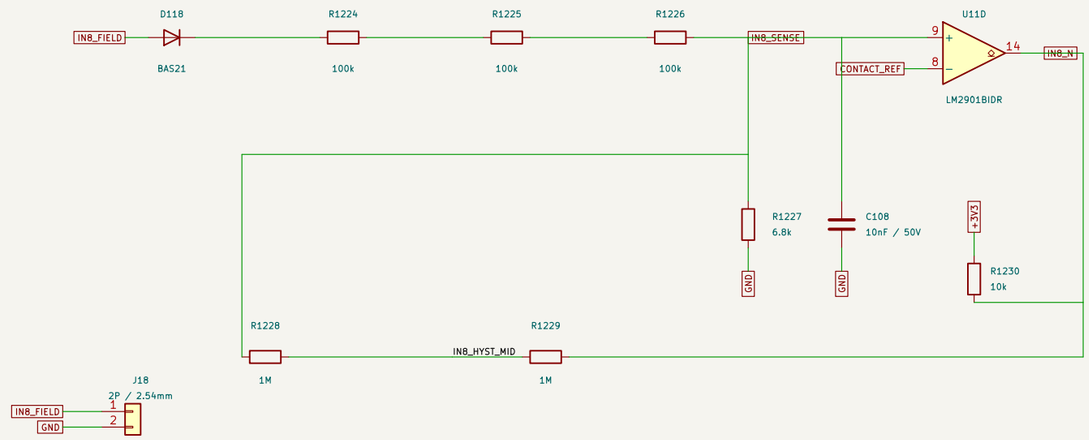
INPUT 7 | ACTIVE-LOW RECEIVER

LOW = closed; FAULT high masks invalid closure



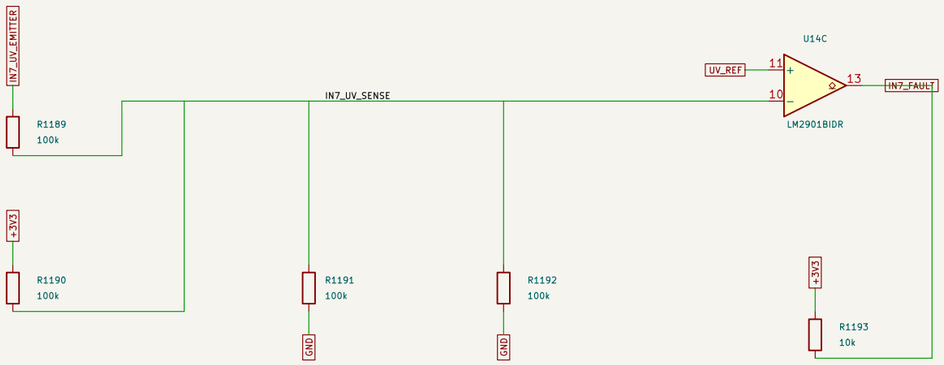
INPUT 8 | ACTIVE-LOW RECEIVER

LOW = closed; FAULT high masks invalid closure



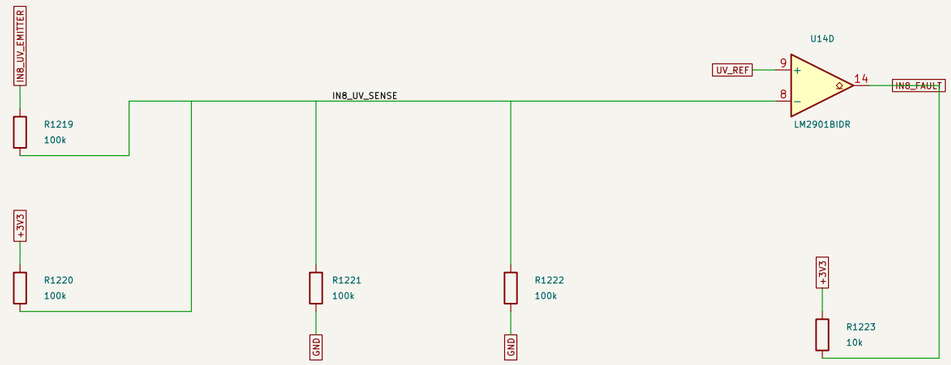
PROTECTED FAULT FLAG

Four 100k conductances bias and attenuate emitter



PROTECTED FAULT FLAG

Four 100k conductances bias and attenuate emitter



Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/-100V faults
Universal Poofer Control | Engineering

Sheet: /10 | Contact receivers 7–8/
File: Input_Receivers_4.kicad_sch

Title: 10 | Contact receivers 7–8

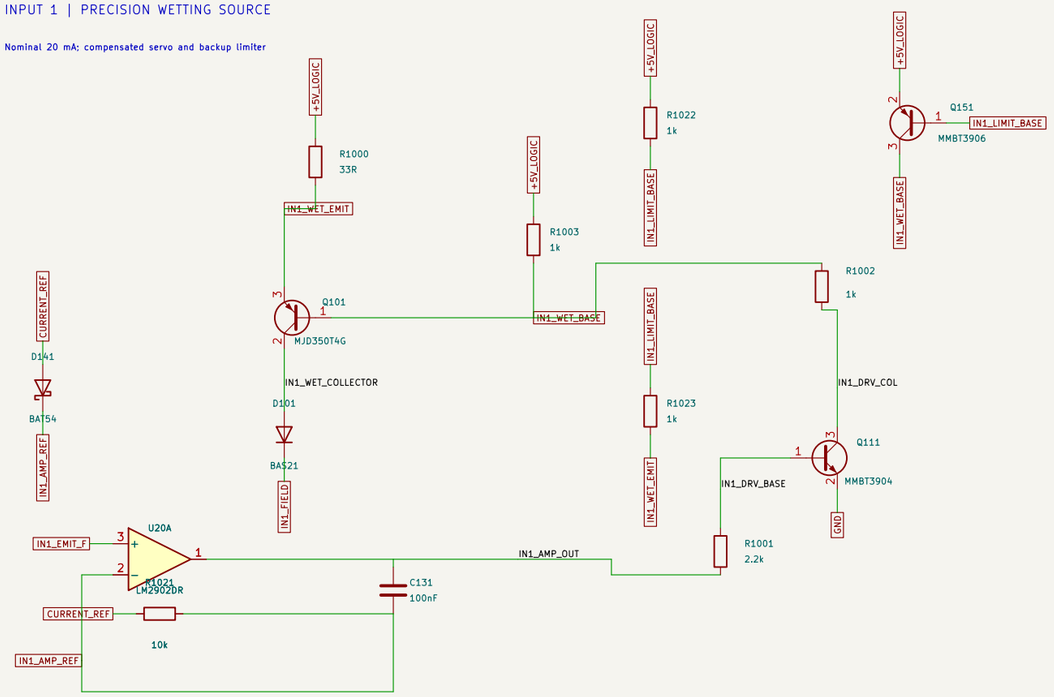
Size: A3	Date: 2026-10-03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 10/24

UNIVERSAL POOFER CONTROL | 11 | Wetting sources 1–2

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

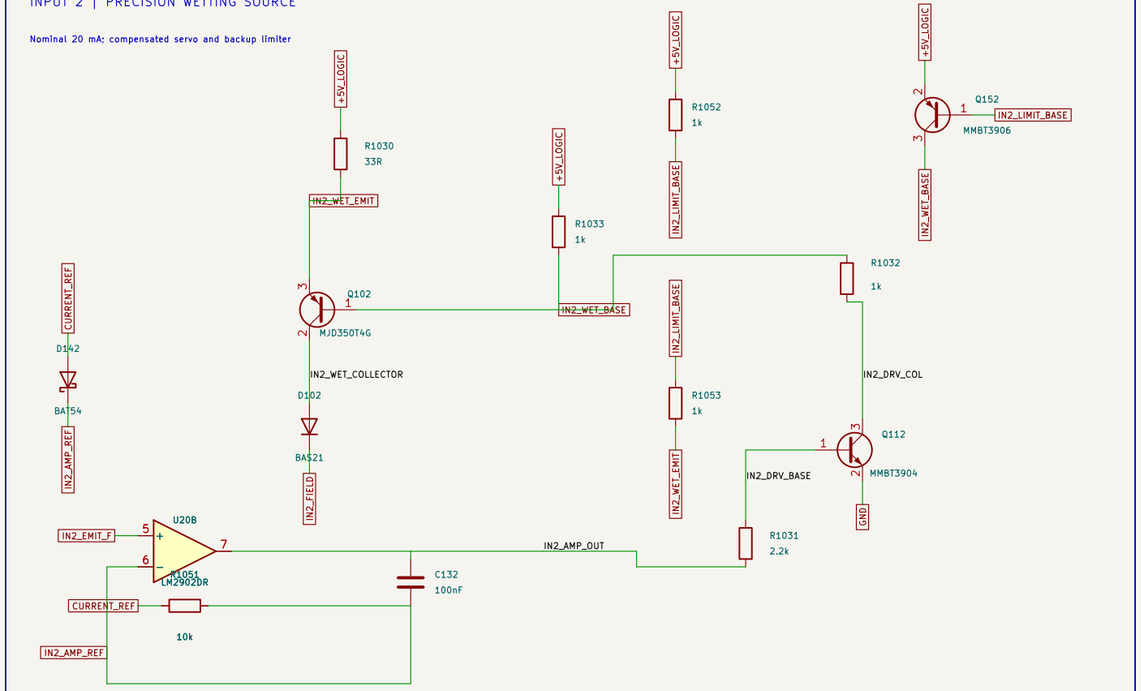
INPUT 1 | PRECISION WETTING SOURCE

Nominal 20 mA; compensated servo and backup limiter

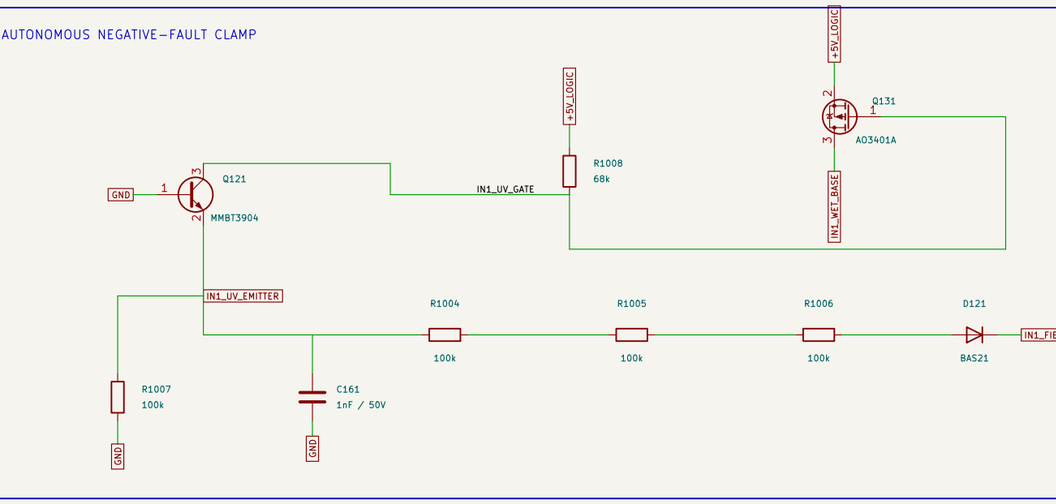


INPUT 2 | PRECISION WETTING SOURCE

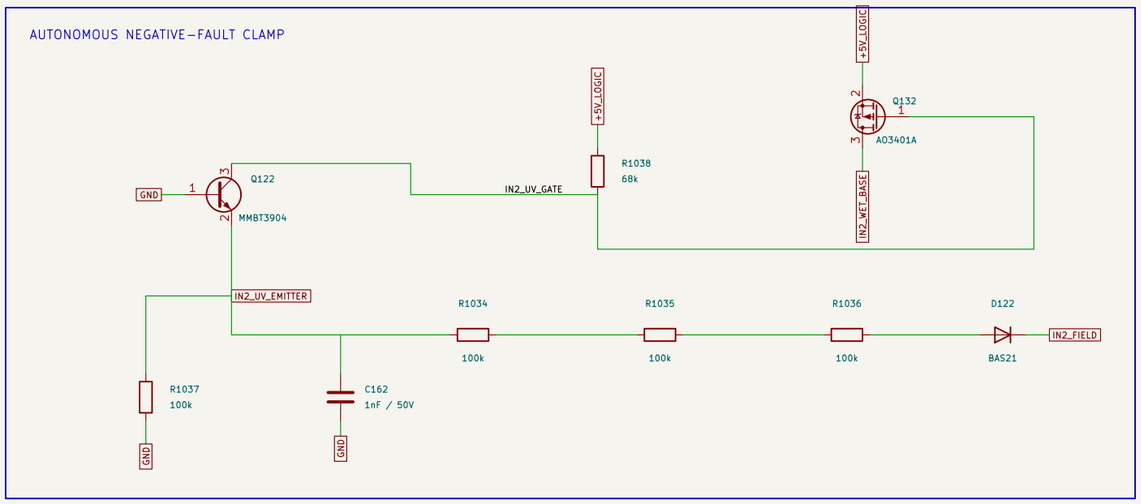
Nominal 20 mA; compensated servo and backup limiter



AUTONOMOUS NEGATIVE–FAULT CLAMP



AUTONOMOUS NEGATIVE–FAULT CLAMP



Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/–100V faults
Universal Poofer Control | Engineering

Sheet: /11 | Wetting sources 1–2/
File: Contact_Sources_1.kicad_sch

Title: 11 | Wetting sources 1–2

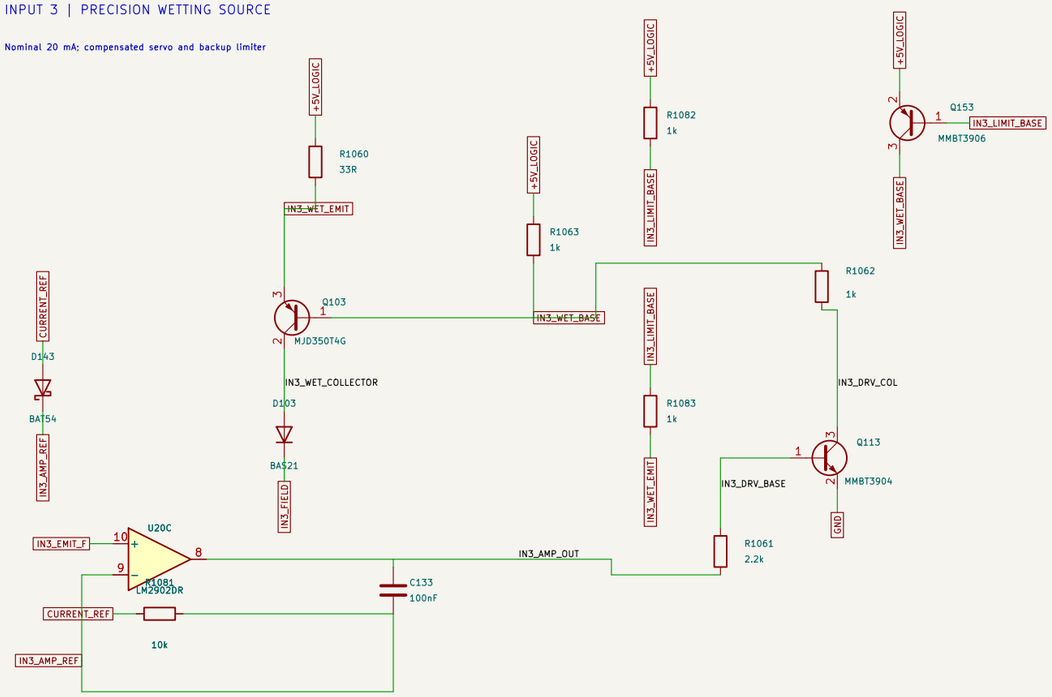
Size: A3	Date: 2026–10–03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 11/24

UNIVERSAL POOFER CONTROL | 12 | Wetting sources 3–4

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

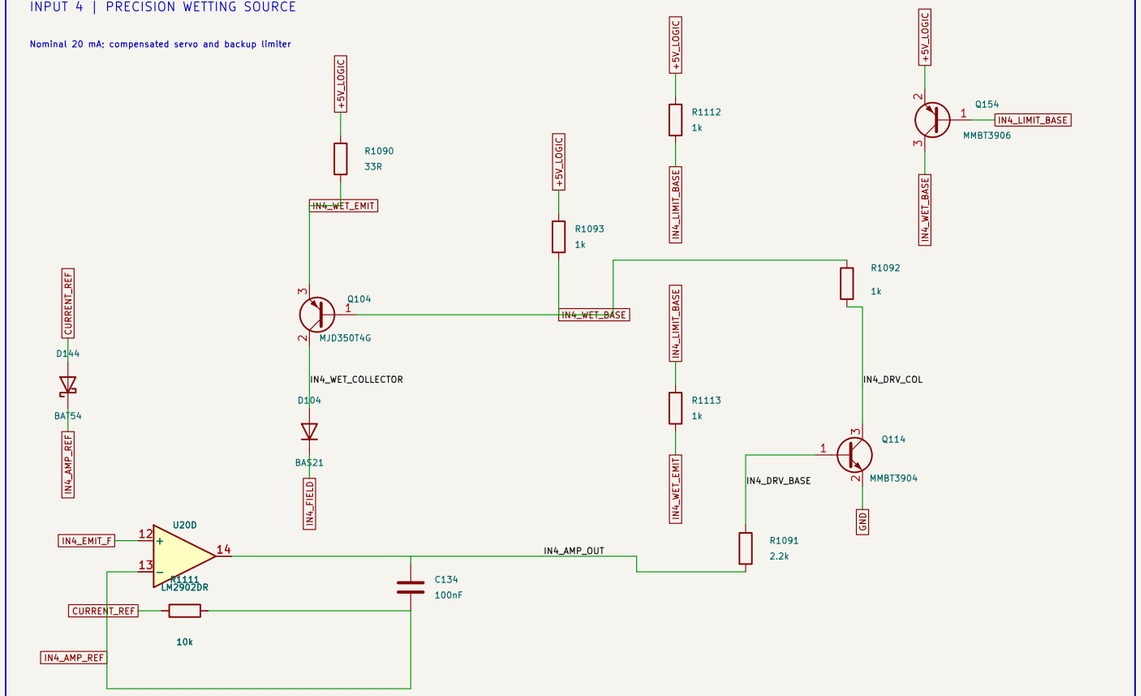
INPUT 3 | PRECISION WETTING SOURCE

Nominal 20 mA; compensated servo and backup limiter

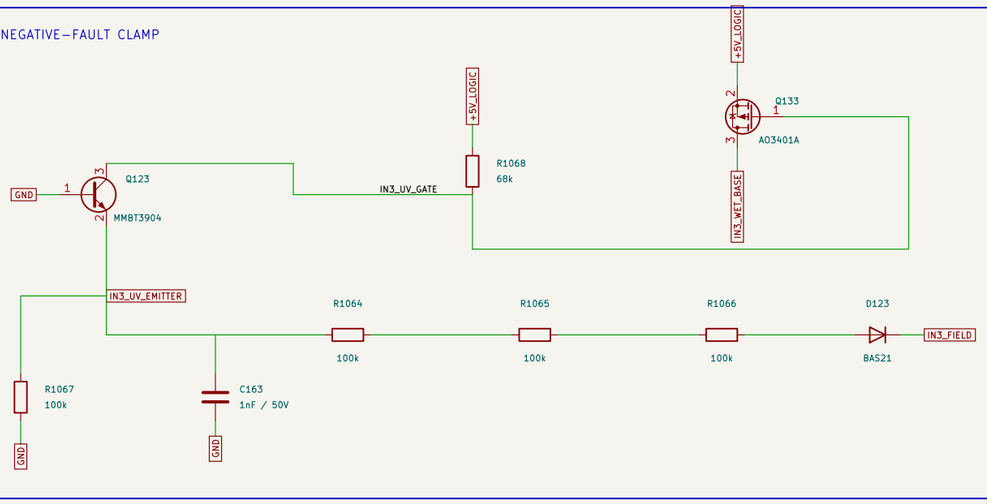


INPUT 4 | PRECISION WETTING SOURCE

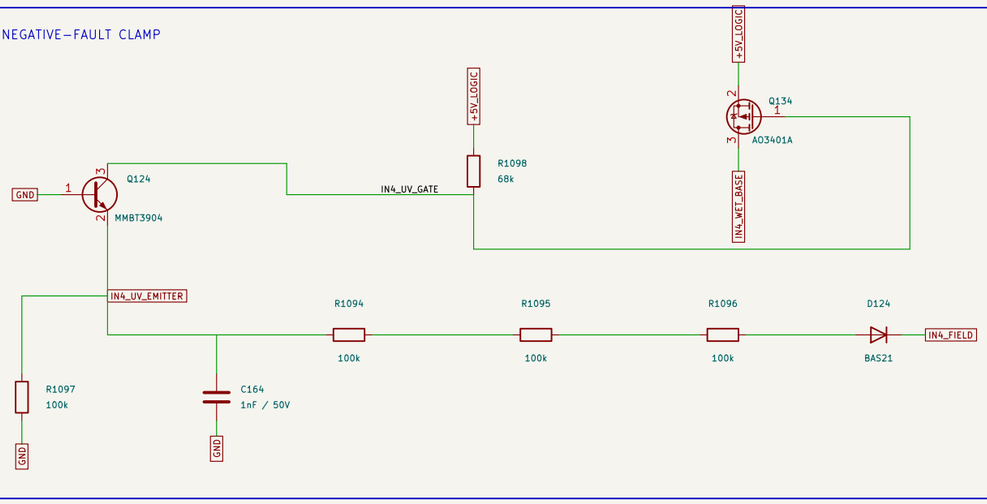
Nominal 20 mA; compensated servo and backup limiter



AUTONOMOUS NEGATIVE–FAULT CLAMP



AUTONOMOUS NEGATIVE–FAULT CLAMP



Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/–100V faults
Universal Poofer Control | Engineering

Sheet: /12 | Wetting sources 3–4/
File: Contact_Sources_2.kicad_sch

Title: 12 | Wetting sources 3–4

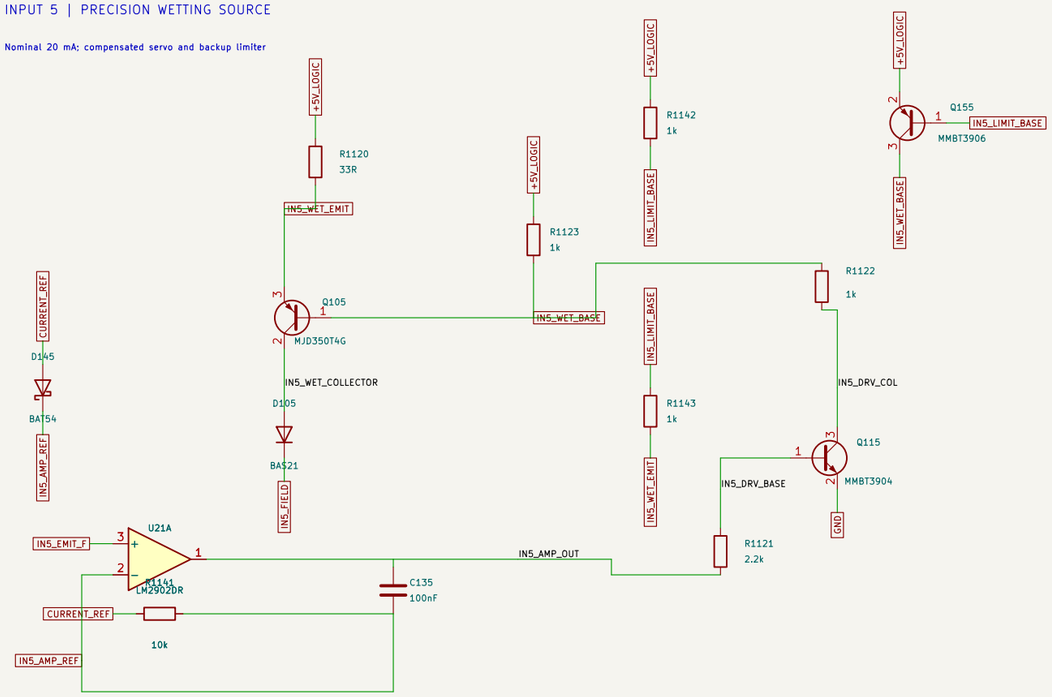
Size: A3	Date: 2026–10–03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 12/24

UNIVERSAL POOFER CONTROL | 13 | Wetting sources 5–6

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

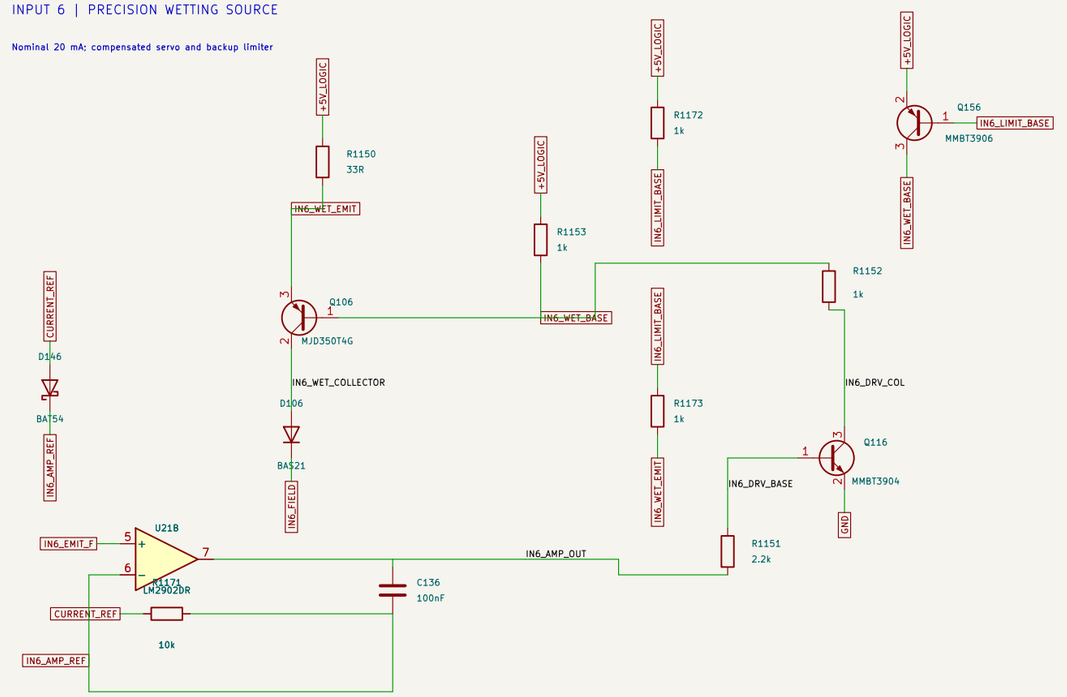
INPUT 5 | PRECISION WETTING SOURCE

Nominal 20 mA; compensated servo and backup limiter

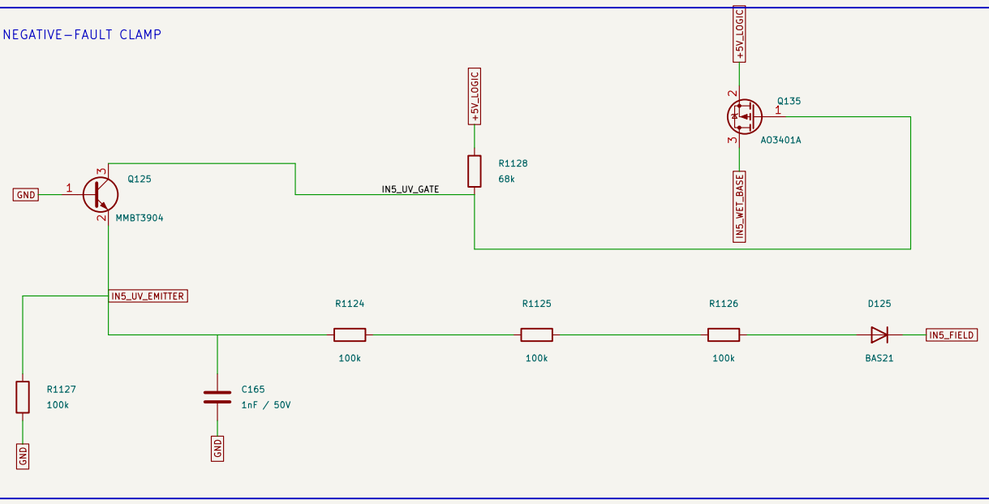


INPUT 6 | PRECISION WETTING SOURCE

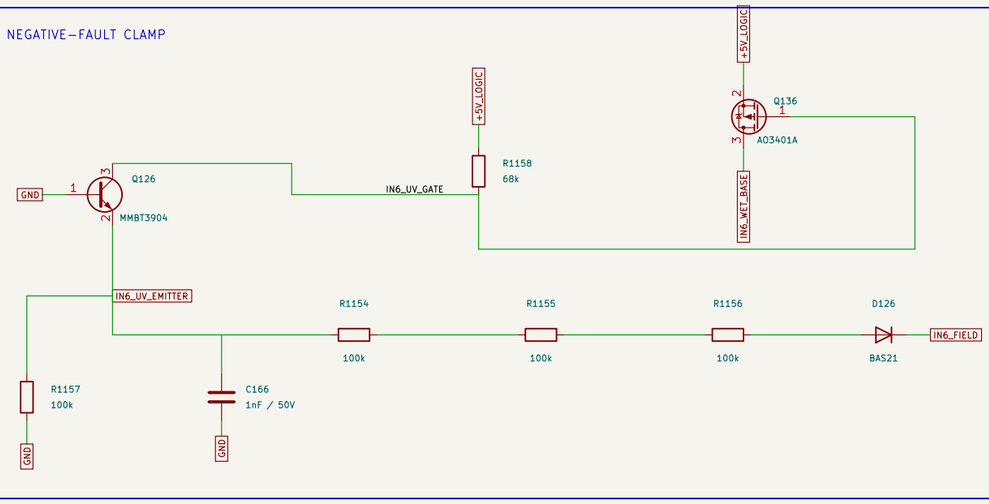
Nominal 20 mA; compensated servo and backup limiter



AUTONOMOUS NEGATIVE–FAULT CLAMP



AUTONOMOUS NEGATIVE–FAULT CLAMP



Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/–100V faults
Universal Poofer Control | Engineering

Sheet: /13 | Wetting sources 5–6/
File: Contact_Sources_3.kicad_sch

Title: 13 | Wetting sources 5–6

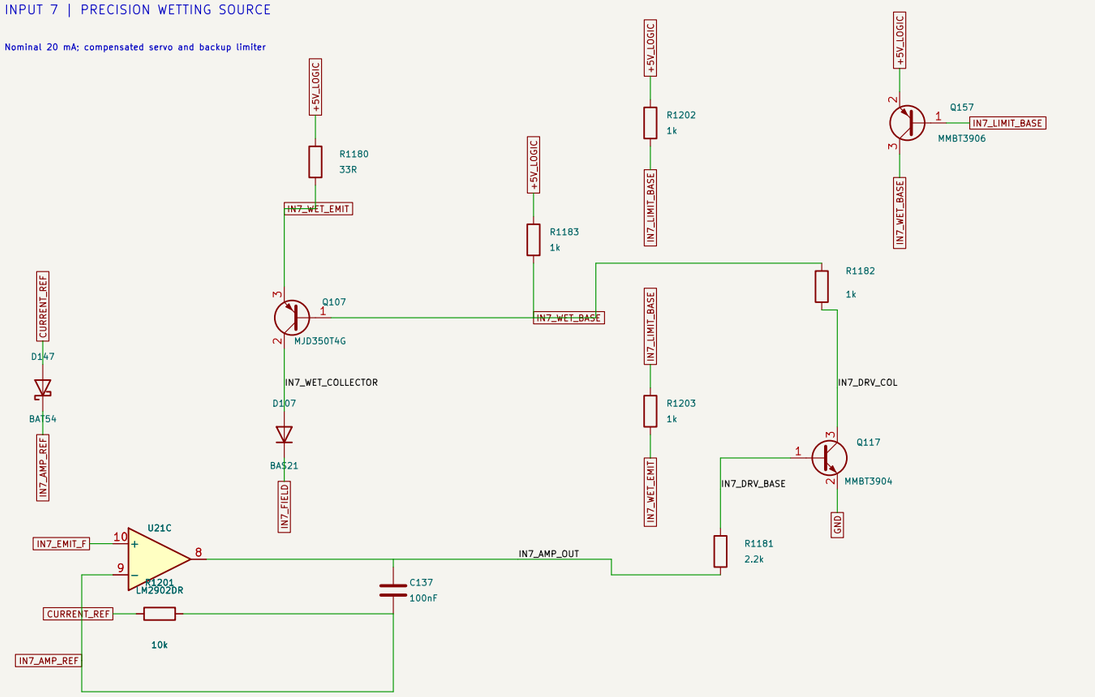
Size: A3	Date: 2026-10-03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 13/24

UNIVERSAL POOFER CONTROL | 14 | Wetting sources 7–8

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

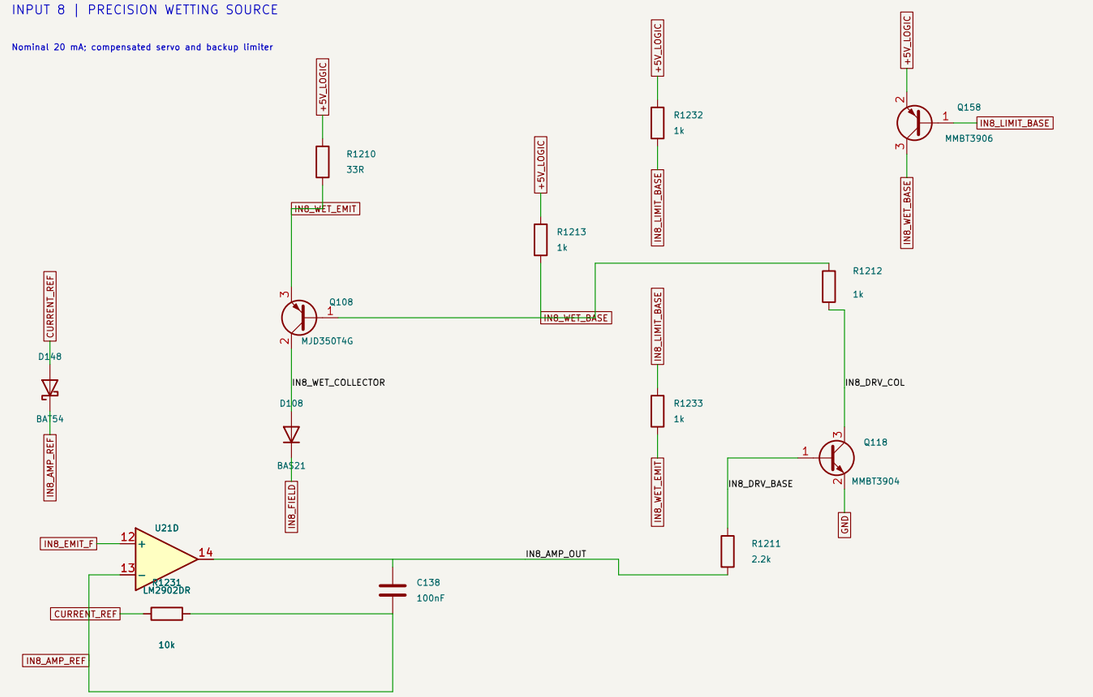
INPUT 7 | PRECISION WETTING SOURCE

Nominal 20 mA; compensated servo and backup limiter

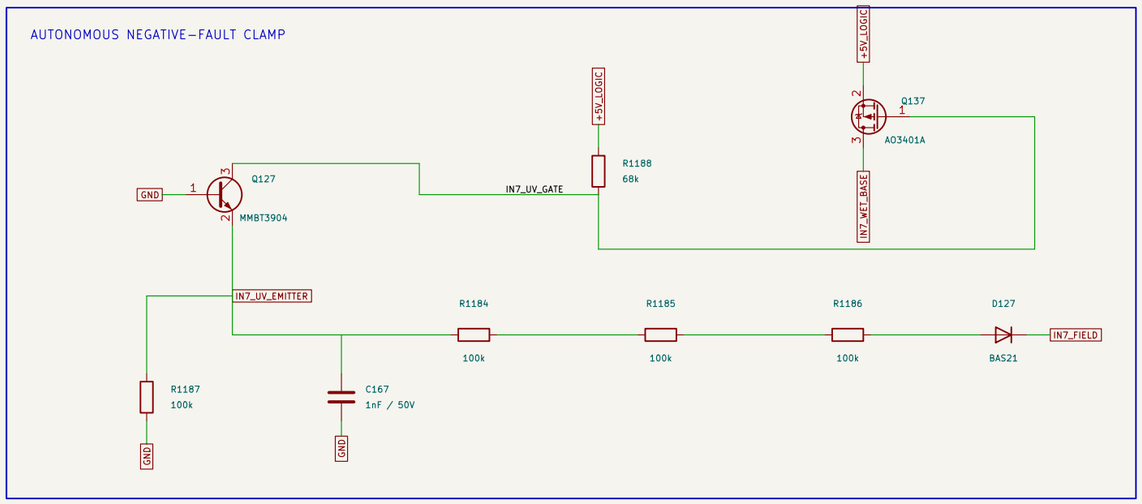


INPUT 8 | PRECISION WETTING SOURCE

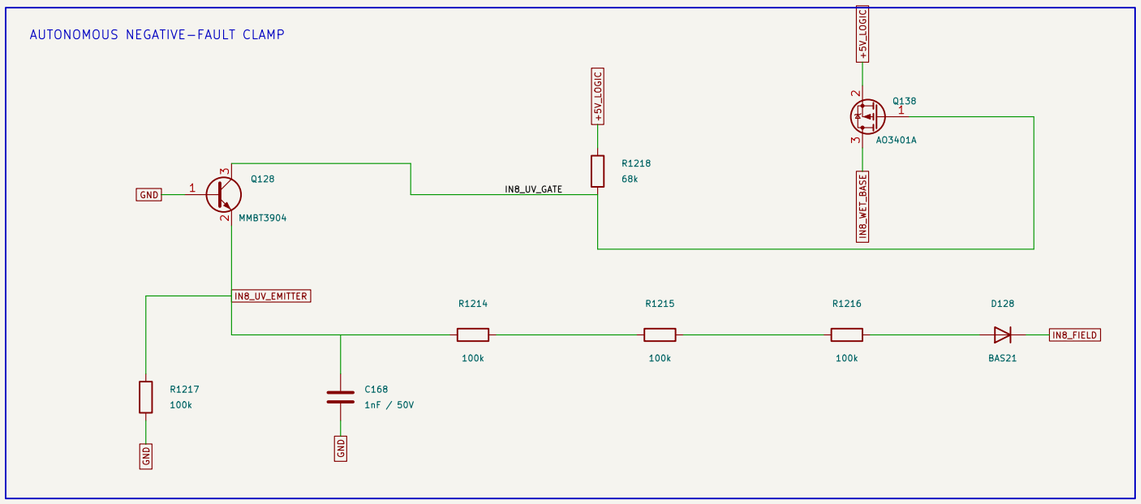
Nominal 20 mA; compensated servo and backup limiter



AUTONOMOUS NEGATIVE–FAULT CLAMP



AUTONOMOUS NEGATIVE–FAULT CLAMP



Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/–100V faults
Universal Poofer Control | Engineering

Sheet: /14 | Wetting sources 7–8/
File: Contact_Sources_4.kicad_sch

Title: 14 | Wetting sources 7–8

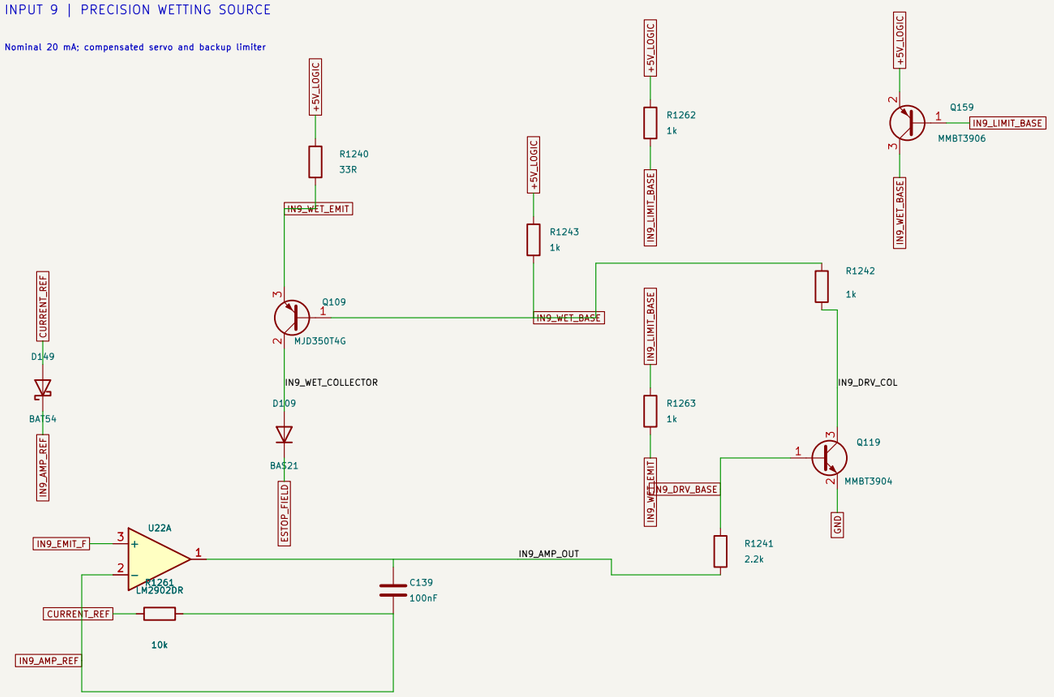
Size: A3	Date: 2026-10-03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 14/24

UNIVERSAL POOFER CONTROL | 15 | NC emergency-stop field loop

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

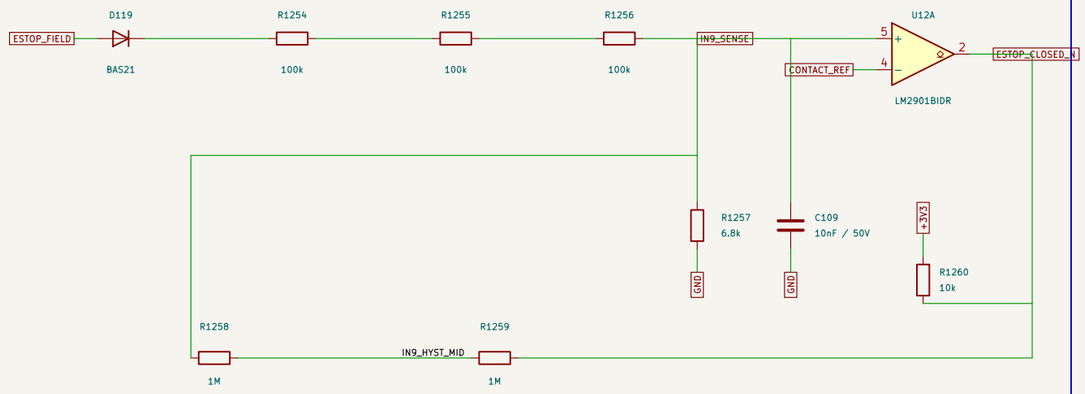
INPUT 9 | PRECISION WETTING SOURCE

Nominal 20 mA; compensated servo and backup limiter

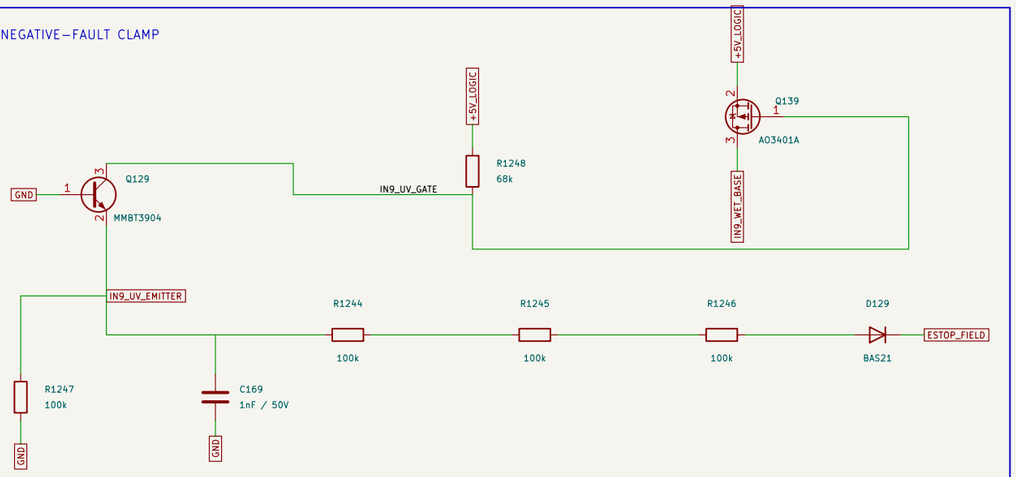


INPUT 9 | ACTIVE-LOW RECEIVER

LOW = closed; FAULT high masks invalid closure

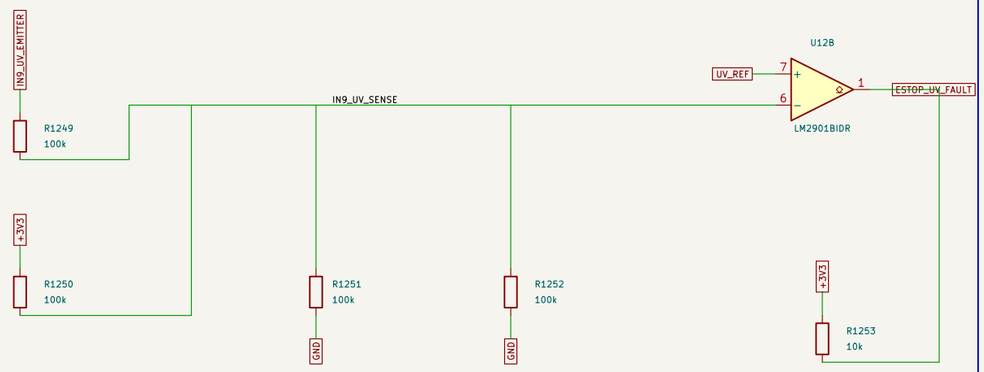


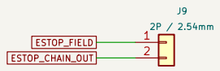
AUTONOMOUS NEGATIVE-FAULT CLAMP

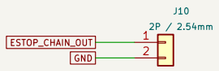


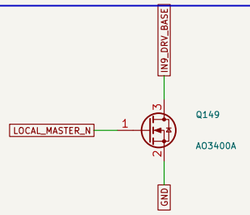
PROTECTED FAULT FLAG

Four 100k conductances bias and attenuate emitter









ExactlyoneMASTERsuppliesloopwetting; all NC contacts and chainconnectors are in series.

Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/-100V faults

Universal Poofer Control | Engineering

Sheet: /15 | NC emergency-stop field loop/
File: Stop_Field.kicad_sch

Title: 15 | NC emergency-stop field loop

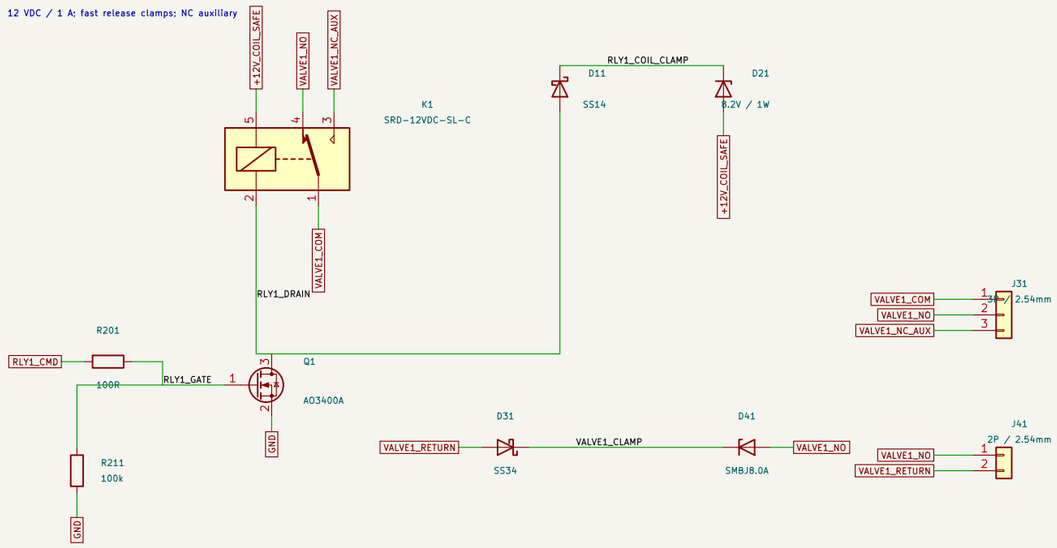
Size: A3	Date: 2026-10-03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 15/24

UNIVERSAL POOFER CONTROL | 16 | Valve outputs 1–4

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

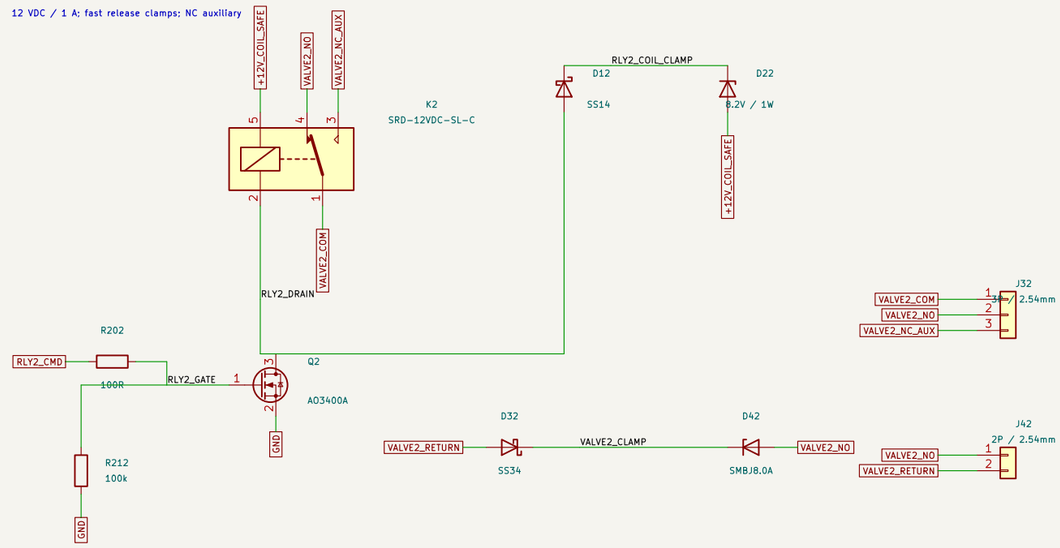
VALVE 1 | NO OUTPUT

12 VDC / 1 A; fast release clamps; NC auxiliary



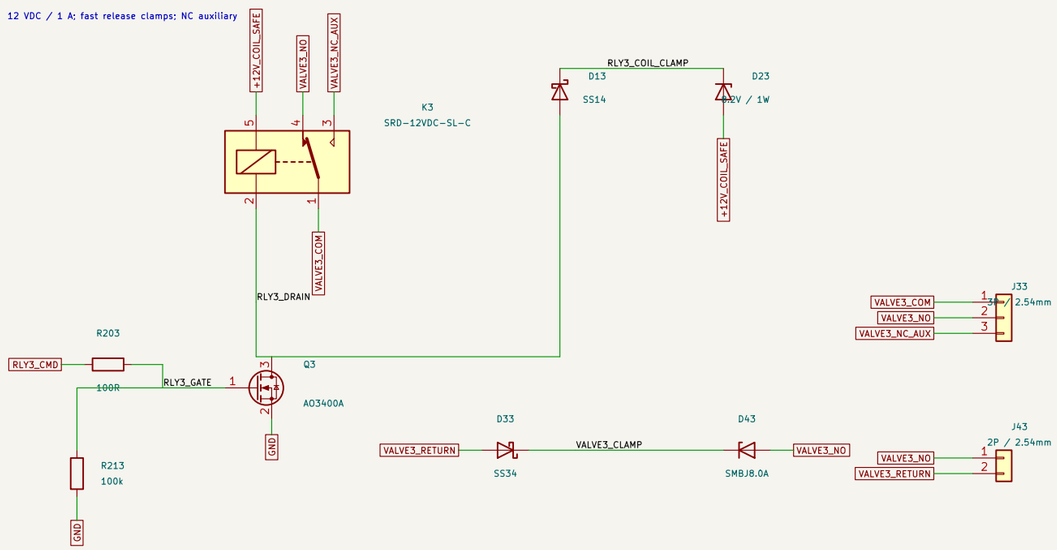
VALVE 2 | NO OUTPUT

12 VDC / 1 A; fast release clamps; NC auxiliary



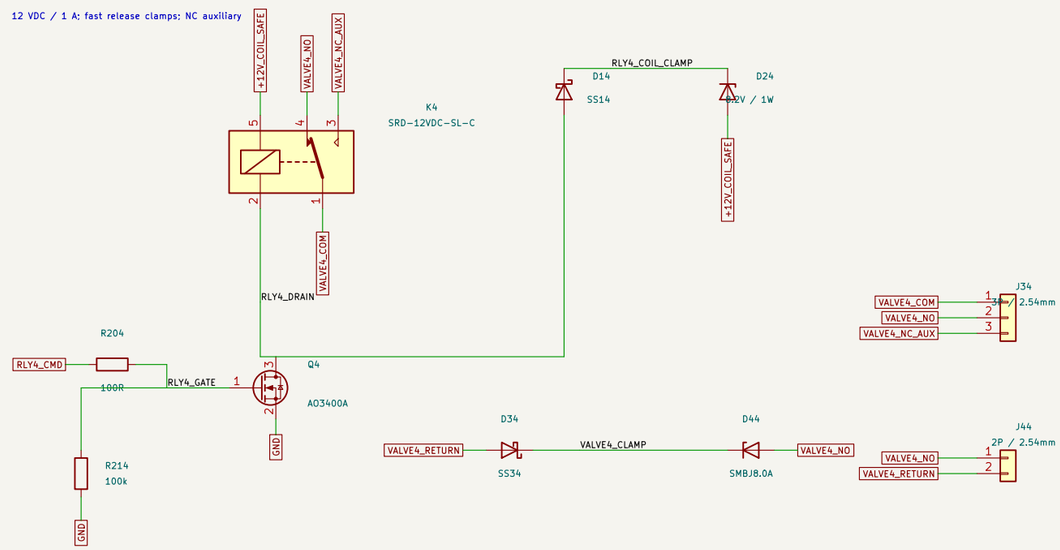
VALVE 3 | NO OUTPUT

12 VDC / 1 A; fast release clamps; NC auxiliary



VALVE 4 | NO OUTPUT

12 VDC / 1 A; fast release clamps; NC auxiliary



Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/–100V faults
Universal Poofer Control | Engineering

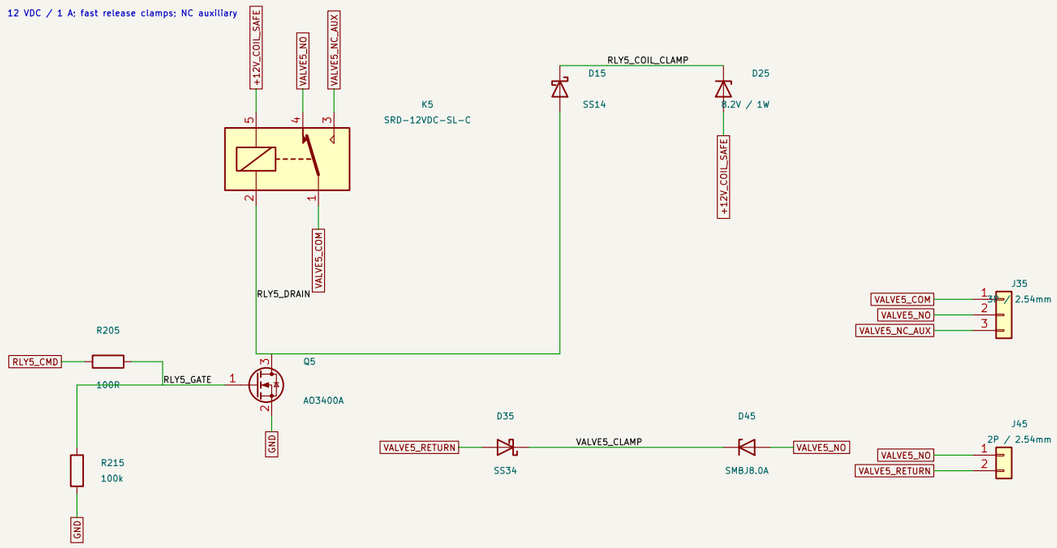
Sheet: /16 | Valve outputs 1–4/		
File: Relay_Outputs_1.kicad_sch		
Title: 16 | Valve outputs 1–4		
Size: A3	Date: 2026-10-03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 16/24

UNIVERSAL POOFER CONTROL | 17 | Valve outputs 5–8

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

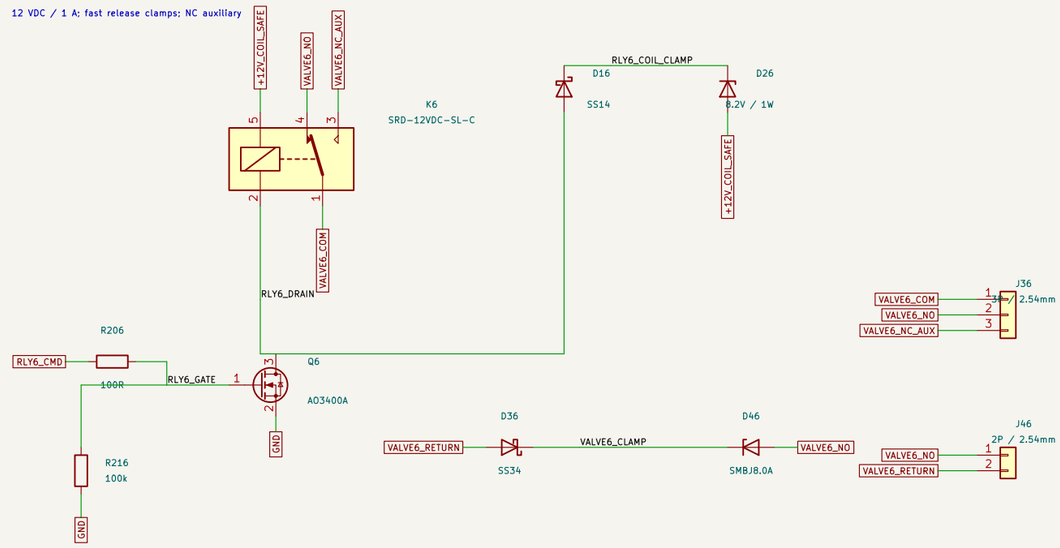
VALVE 5 | NO OUTPUT

12 VDC / 1 A; fast release clamps; NC auxiliary



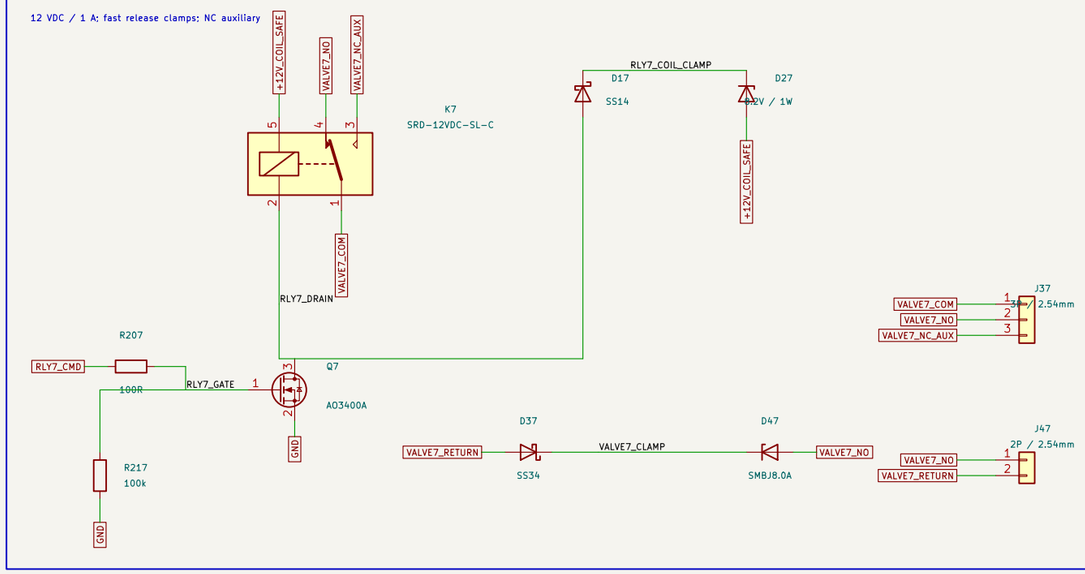
VALVE 6 | NO OUTPUT

12 VDC / 1 A; fast release clamps; NC auxiliary



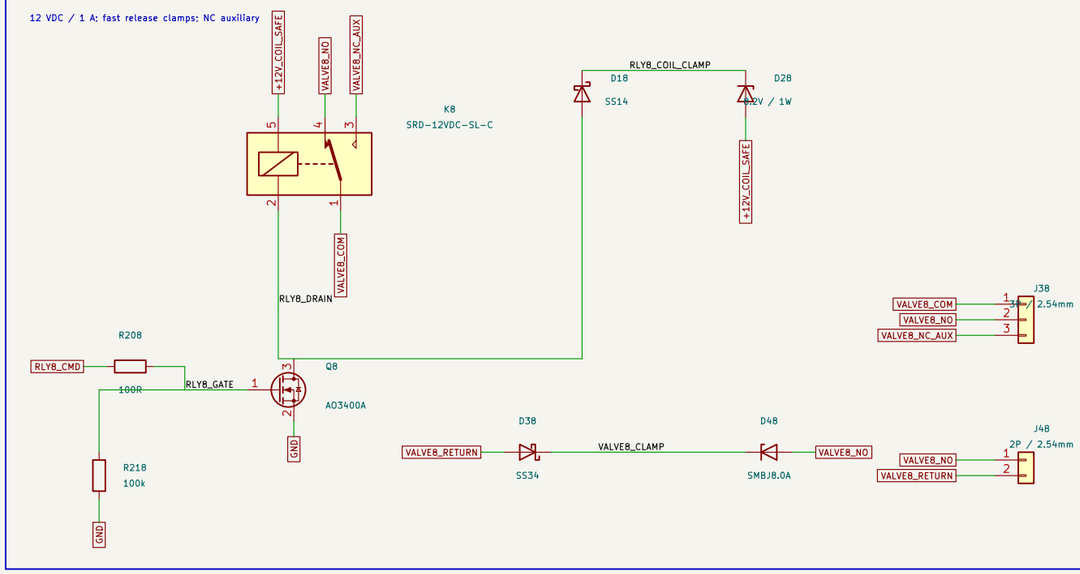
VALVE 7 | NO OUTPUT

12 VDC / 1 A; fast release clamps; NC auxiliary



VALVE 8 | NO OUTPUT

12 VDC / 1 A; fast release clamps; NC auxiliary



Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/−100V faults
Universal Poofer Control | Engineering

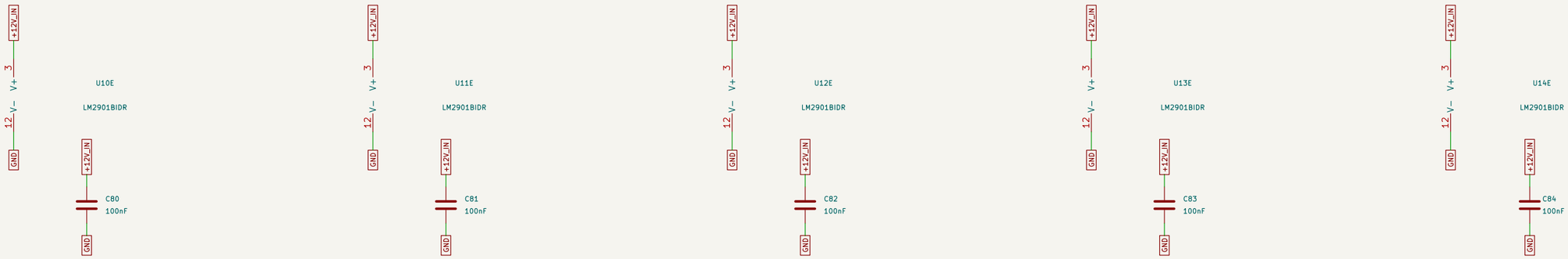
Sheet: /17 | Valve outputs 5–8/		
File: Relay_Outputs_2.kicad_sch		
Title: 17 | Valve outputs 5–8		
Size: A3	Date: 2026-10-03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 17/24

UNIVERSAL POOFER CONTROL | 18 | Analog supplies, spare units and debug port

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

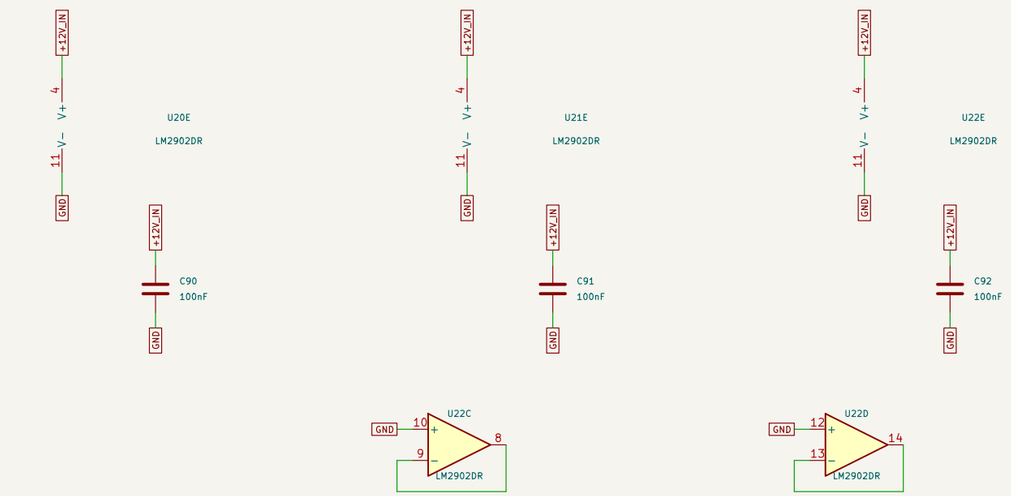
COMPARATOR BANK SUPPLIES

One 100 nF bypass per physical IC; raw12V supply



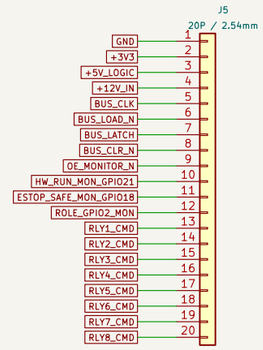
CURRENT-SERVO SUPPLIES

Defined unused amplifiers; no floating inputs

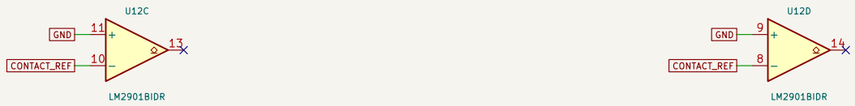


DEBUG / COMMAND BREAKOUT

Power pins identified; monitor nodes isolated



DEFINED UNUSED COMPARATORS



Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/−100V faults
Universal Poofer Control | Engineering

Sheet: /18 | Analog supplies, spare units and debug port/
File: Analog_Support.kicad_sch

Title: 18 | Analog supplies, spare units and debug port

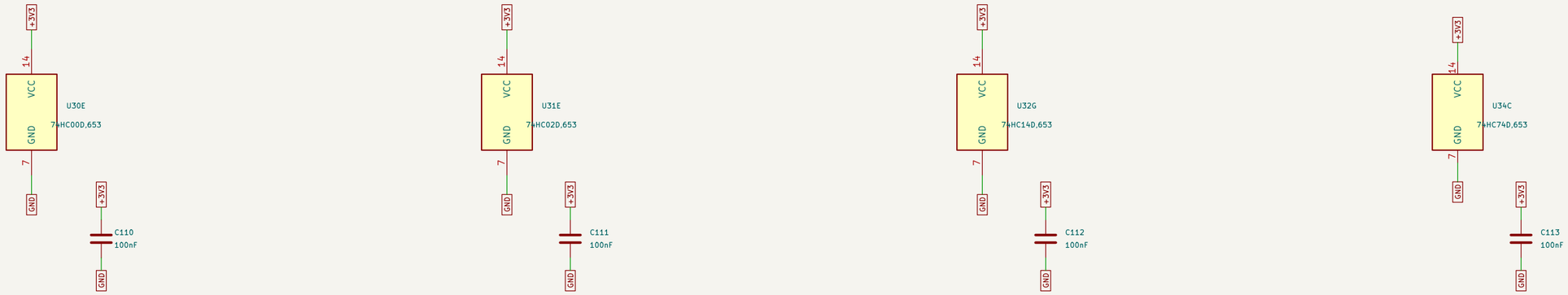
Size: A3	Date: 2026-10-03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 18/24

UNIVERSAL POOFER CONTROL | 19 | Safety logic supplies and unused gates

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

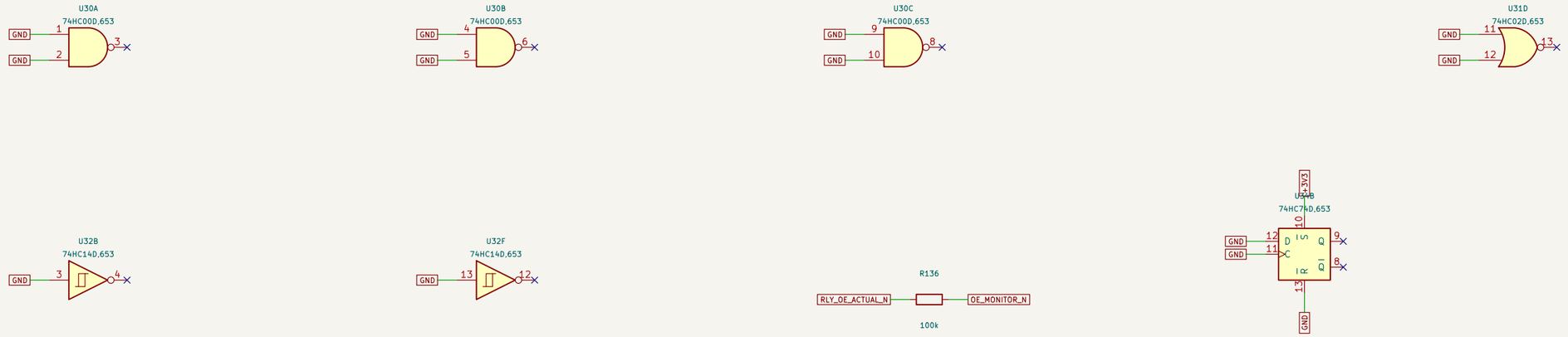
SAFETY LOGIC SUPPLIES

Local 3.3V rail; 100nF at each package



DEFINED SPARE GATES

No floating CMOS inputs; unused outputs left open



Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/−100V faults
Universal Poofer Control | Engineering

Sheet: /19 | Safety logic supplies and unused gates/
File: Logic_Support.kicad_sch

Title: 19 | Safety logic supplies and unused gates

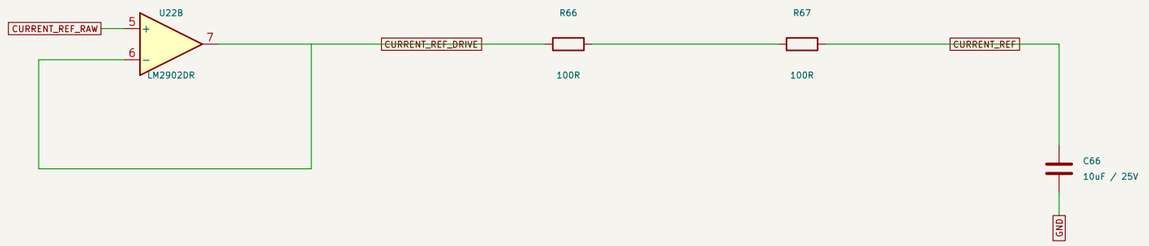
Size: A3	Date: 2026-10-03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 19/24

UNIVERSAL POOFER CONTROL | 20 | Buffered current reference and rail sequencing

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

LOW-IMPEDANCE SHARED REFERENCE

SpareU22B isolates all nine compensated source loops



200ohm total bounds initial10uF discharge near22mA. Output behavior whenunpowered requireshardwarequalification.

Per-source10k/100nF compensation uses a private negative-input node; never attach it directly to the rawdivider.

Backup PNP limiters bound saturation recovery current. Power-good holds colloutputs off through settling.

Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/-100V faults
Universal Poofer Control | Engineering

Sheet: /20 | Buffered current reference and rail sequencing/
File: Current_Reference.kicad_sch

Title: 20 | Buffered current reference and rail sequencing

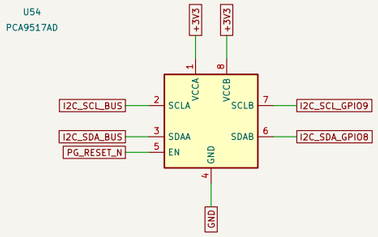
Size: A3	Date: 2026-10-03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 20/24

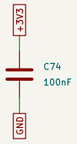
UNIVERSAL POOFER CONTROL | 21 | I2C expansion and power-off isolation

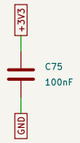
REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

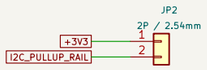
COMMON A BUS / PRIVATE B BUS

All A sides may share the bus; B sides must remain private



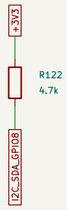


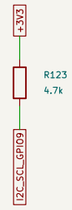












Pins15SDA,16SCL on both chain ports. Default 100 kHz; qualify totalcommon-A capacitance<=400pF.

32 A ports can contribute224pF before cabling.2.2k commonpullups meet the conservativeleakage/high-levelbudget.

MCU B-side LOW mustbe<=0.4V; bufferedLOW<=0.6V remainsbelowESP VIL. Testload/temperaturecorners.

Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/-100V faults
Universal Poofer Control | Engineering

Sheet: /21 | I2C expansion and power-off isolation/
File: I2C_Expansion.kicad_sch

Title: 21 | I2C expansion and power-off isolation

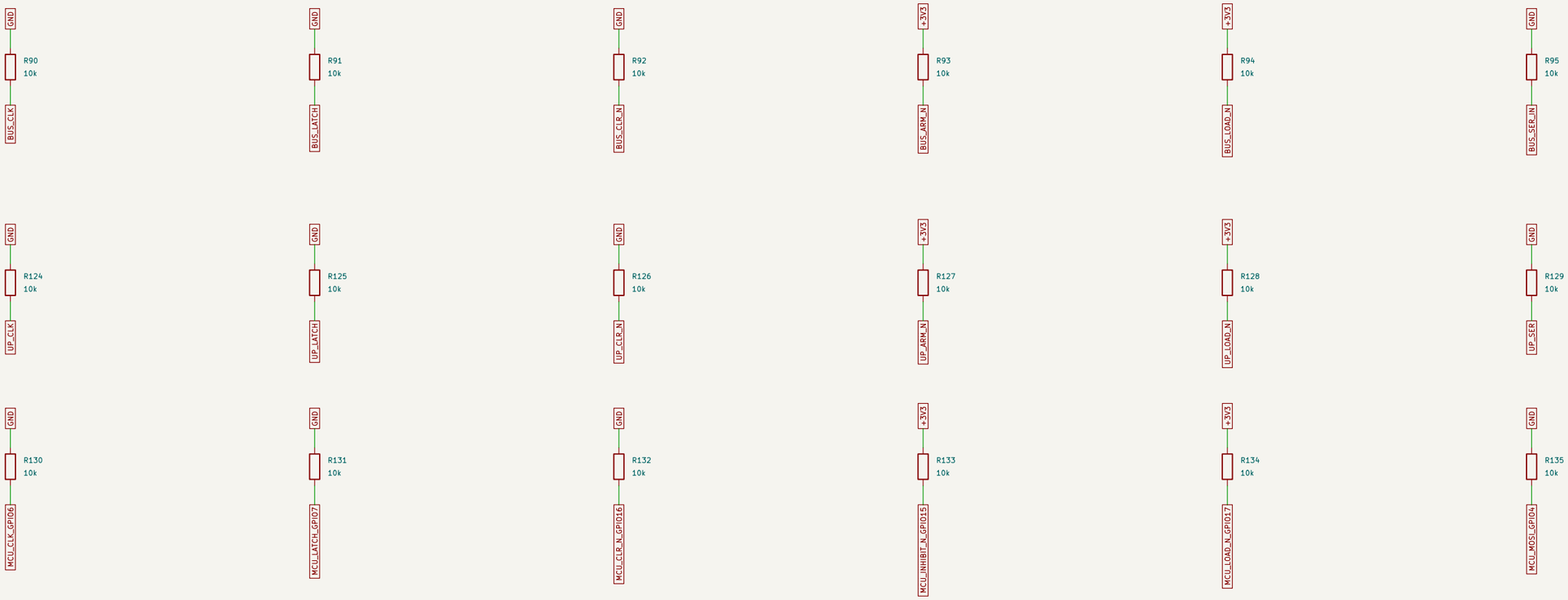
Size: A3	Date: 2026-10-03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 21/24

UNIVERSAL POOFER CONTROL | 22 | Expansion defaults and buffer supplies

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

DEFINED COMMAND DEFAULTS

BUS / UP / LOCAL rows: safe receiver inputs even if MCU never boots or cable is absent



CHAIN TAIL / LOCAL BYPASS

Tail HIGH fills missing-module fault bits with FF; firmware masks these input bytes



Keep each 100nF physically beside its LVC244 VCC pin; no supply is shared by the expansion connectors.

Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/-100V faults
Universal Poofer Control | Engineering

Sheet: /22 | Expansion defaults and buffer supplies/
File: Expansion_Bias.kicad_sch

Title: 22 | Expansion defaults and buffer supplies

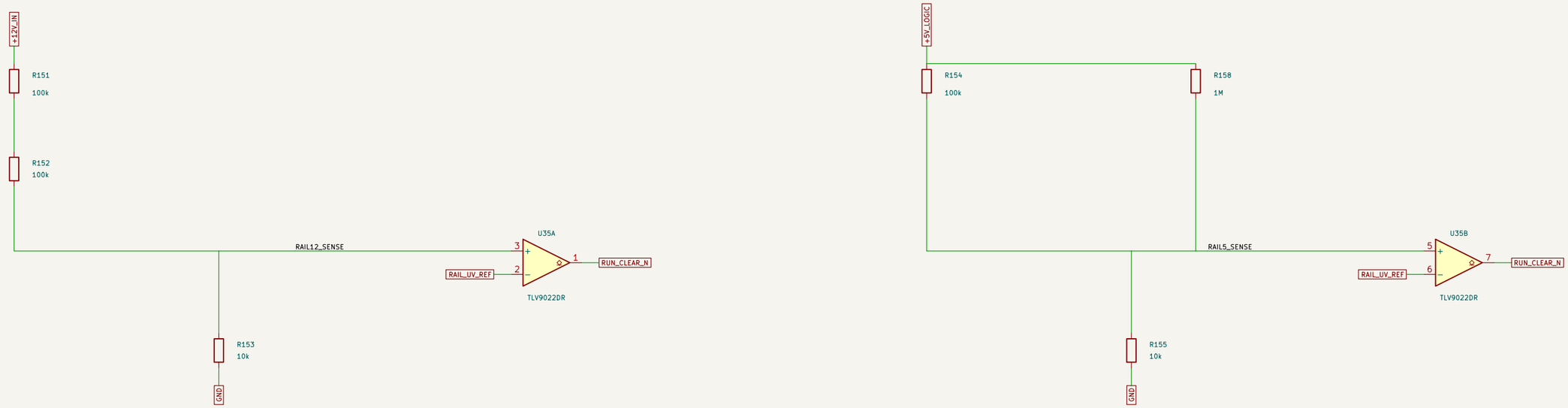
Size: A3	Date: 2026-10-03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 22/24

UNIVERSAL POOFER CONTROL | 23 | Hardware monitoring of 5V and raw 12V

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

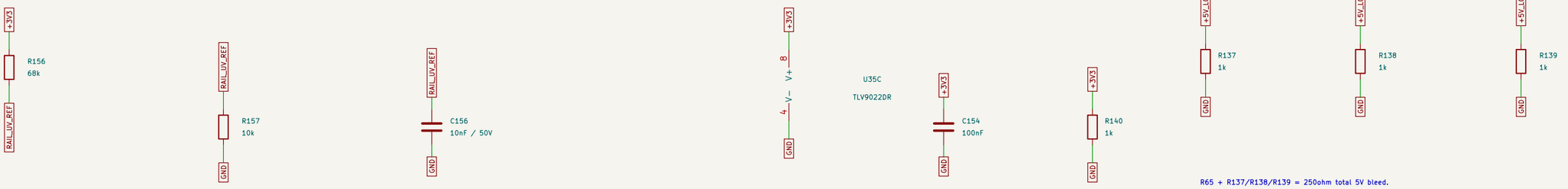
LOGIC-POWERED SUPPLY LOSS DETECTION

Asynchronous RUN clear remains powered when the coil and analog rail disappears



COMMON REFERENCE / LOCAL SUPPLY

5V trip about4.27V;12V trip about8.88V: loss and restoration cannot clock RUN



R65 + R137/R138/R139 = 250ohm total 5V bleed.

Comparator common-mode inputs remain below0.7V, including13.2V raw power, while the3.3V supervisor is valid.

Open-drain outputs sink flip-flop clear;4.7k limits HC02 source current. Real rail ramps and recovery require tests.

Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/-100V faults
Universal Poofer Control | Engineering

Sheet: /23 | Hardware monitoring of 5V and raw 12V/
File: Rail_Supervision.kicad_sch

Title: 23 | Hardware monitoring of 5V and raw 12V

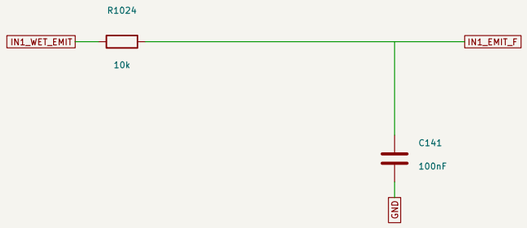
Size: A3	Date: 2026-10-03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 23/24

UNIVERSAL POOFER CONTROL | 24 | Stable emitter feedback for all wetting sources

REV A.3 | Dry contacts | Latched hardware stop | Buffered expansion

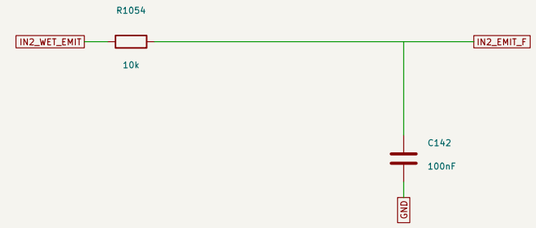
INPUT 1 | EMITTER FEEDBACK

10k / 100n low-pass removes high-frequency plant gain



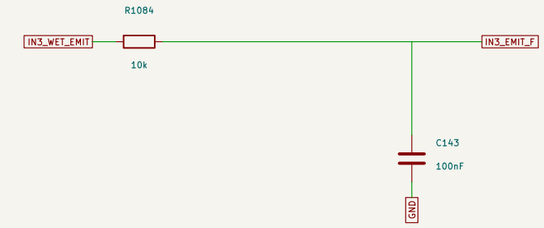
INPUT 2 | EMITTER FEEDBACK

10k / 100n low-pass removes high-frequency plant gain



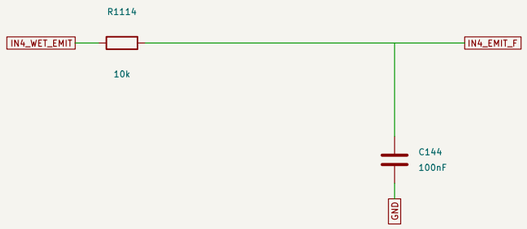
INPUT 3 | EMITTER FEEDBACK

10k / 100n low-pass removes high-frequency plant gain



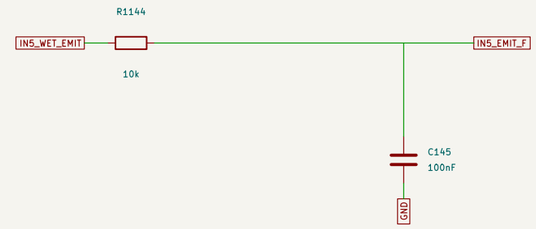
INPUT 4 | EMITTER FEEDBACK

10k / 100n low-pass removes high-frequency plant gain



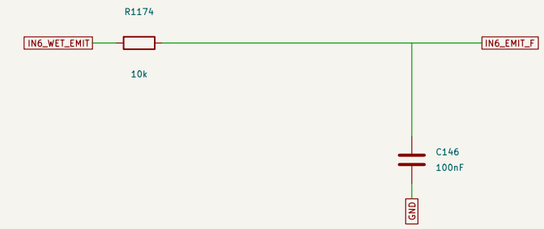
INPUT 5 | EMITTER FEEDBACK

10k / 100n low-pass removes high-frequency plant gain



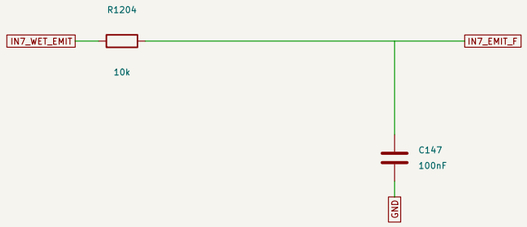
INPUT 6 | EMITTER FEEDBACK

10k / 100n low-pass removes high-frequency plant gain



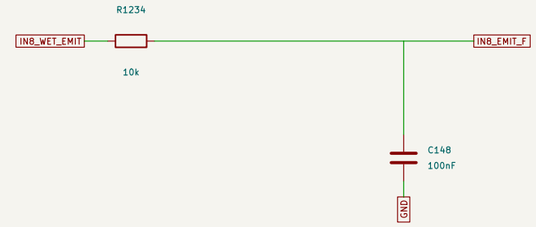
INPUT 7 | EMITTER FEEDBACK

10k / 100n low-pass removes high-frequency plant gain



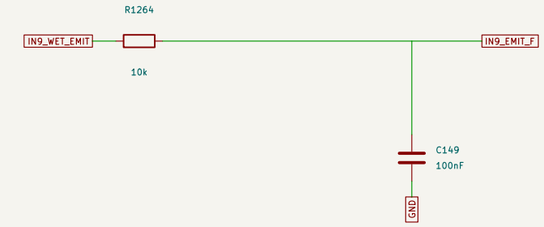
INPUT 8 | EMITTER FEEDBACK

10k / 100n low-pass removes high-frequency plant gain



INPUT 9 | EMITTER FEEDBACK

10k / 100n low-pass removes high-frequency plant gain



Retain the separate OUT-to-IN- compensation. These filters improve illustrative AC phase margin; real parts still require tests.

Hardware stop latch | 12V / 1A NO valve outputs
Grounded contacts / 20mA wetting / +/-100V faults
Universal Poofer Control | Engineering

Sheet: /24 | Stable emitter feedback for all wetting sources/
File: Source_Feedback_Filters.kicad_sch

Title: 24 | Stable emitter feedback for all wetting sources

Size: A3	Date: 2026-10-03	**Rev: A.3**
KiCad E.D.A. 10.0.5		Id: 24/24